Econometrics of Commodity and Asset Pricing

Lecture 4

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ENSAE, Finance & Gestion des Risques, MiE, 2025 - 2026

Agenda - Lecture 4

I. A New Perspective on Gaussian ATSMs - Part II.

II. Switching Regimes Gaussian ATSMs

III. Non-Negative ATSMs

Chapter I

A NEW PERSPECTIVE ON GATSMs
- PART II -

– Agenda –

I.1 – INTRODUCTION

- **Gaussian ATSMs typically feature the following ingredients**:

(*i*) a vector of state variables (risk factors) following a Gaussian VAR process (under $\mathbb{P}$ and $\mathbb{Q}$);

(*ii*) a low-dimensional factor-structure in which the risk factors are both macroeconomic and yield-based variables;

(*iii*) the assumption of no-arbitrage opportunities in bond markets.

(*iv*) a risk-free short rate affine function of the state variables.

- **As far as the second condition is concerned**:
  - the low dimension is motivated by the small (two or three) number of principal components (PCs) required the variation of bond yields.
  - Macro variables are typically economic activity and inflation rate.

- **Under the previous conditions**:
  - theoretical yields are affine functions of *all* state variables;
  - this class of macro-finance Gaussian ATSMs is denoted *MTSM*.

I.1 – INTRODUCTION

☐ **Notations**:

- yields are collected in the vector Y_t which contains rates for J different maturities;
- the risk factors are denoted Z_t and include yield-based factors and macro variables;
- we denote the $\mathcal{M}$ macro factors by M_t.

☐ **Regarding yield-based factors**:

- we are free to choose any specific yields or linear combination of yields;
- we write W for a $(J \times J)$ full-rank matrix that defines "portfolios" (linear combinations) of yields: $P_t = W\, Y_t$;
- we denote P_t^j and W^j the first j yield portfolios and their weights;
- we take the first $\mathcal{L}$ PCs (linear combination of yields), $P_t^{\mathcal{L}}$, as the yield factors and their loadings make up the rows of W.

☐ **The vector of risk factors Z_t**:

- is of dimension $\mathcal{N} = \mathcal{L} + \mathcal{M}$ and given by $Z_t = (P_t^{\mathcal{L}\prime}, M_t')'$.

I.2 – A CANONICAL GATSM

☐ **A canonical** ***MTSM***:

- is maximally flexible, (i.e., each member of the family of *MTSMs* is represented) and enforces a minimal set of normalization conditions to ensure econometric identification;
- provides an organizing framework within which it is possible to (analytically) impose the relevant ICCs on the selected PCs;
- and leads directly to a formal characterization of the added flexibility of a *MTSMs* relative to an $\mathcal{N}$-factor model with no macro variables.

☐ **The risk-free short rate**: $r_t = \rho_0 + \rho'_{1M}\, M_t + \rho'_{1P}\, P_t^{\mathcal{L}} = \rho_0 + \rho'_1\, Z_t$;

- some treat $P_t^{\mathcal{L}}$ as a set of latent risk factors, while others consider them as (observable) portfolio of yields; we will see that these two theoretical formulations are observationally equivalent.

☐ **In fact, we will see that**:

- we are free to rotate the entire vector Z_t to express bond prices in terms of $P_t^{\mathcal{N}}$, the first $\mathcal{N} = \mathcal{L} + \mathcal{M}$ entries of portfolios of yields P_t ($\mathcal{N} \leq J$);
- this is an implication of the affine pricing of $P_t^{\mathcal{N}}$ in terms of Z_t.
- Accordingly, we are free to start with either interpretation of $P_t^{\mathcal{L}}$ (latent or yield-based) and to use any of these rotations of the risk factors Z_t.

I.2 – A CANONICAL GATSM

□ **We assume**:

$$\begin{aligned} \Delta Z_t &= K_0^{\mathbb{Q}} + K_1^{\mathbb{Q}} Z_{t-1} + \sqrt{\Sigma}\,\varepsilon_t^{\mathbb{Q}}, \;\; \varepsilon_t^{\mathbb{Q}} \sim N(0, I)\,, \\ r_t &= \rho_0 + \rho_{1M}' M_t + \rho_{1P}' P_t^{\mathcal{L}} = \rho_0 + \rho_1' Z_t \end{aligned} \tag{I.1}$$

and under AAO we obtain:

$$Y_t = A + B\,Z_t = A + B_P\,P_t^{\mathcal{L}} + B_M\,M_t\,. \tag{I.2}$$

- If we premultiply by W or $W^{\mathcal{N}}$ the previous equation we obtain:

$$\begin{aligned} P_t &= W\,A + W\,B\,Z_t = A_{TS} + B_{TS}\,Z_t \\ P_t^{\mathcal{N}} &= W^{\mathcal{N}}\,A + W^{\mathcal{N}}\,B\,Z_t = A_{TS}^{\mathcal{N}} + B_{TS}^{\mathcal{N}}\,Z_t \end{aligned} \tag{I.3}$$

where the loadings (A_{TS}, B_{TS}) are known functions of $\mathbb{Q}$-parameters $(K_0^{\mathbb{Q}}, K_1^{\mathbb{Q}}, \rho_0, \rho_1, \Sigma)$.

□ **A canonical version** is obtained by imposing normalizations such that the only admissible rotation of Z_t that leaves the distribution of r_t unaffected is the identity matrix.

I.2 – A CANONICAL GATSM

- After rotating to a pricing model with risk factors $P_t^{\mathcal{N}}$, we adopt the canonical form of Joslin, Singleton and Zhu (2011, RFS). **This canonical form** is such that the $\mathbb{Q}$-distribution of $P_t^{\mathcal{N}}$ is fully characterized by:

a) the historical conditional variance-covariance matrix Σ,

b) the marginal mean $r_\infty^{\mathbb{Q}} = \mathbb{E}^{\mathbb{Q}}(r_t)$ of r_t,

c) the rotation-invariant $\mathcal{N}$-vector $\lambda^{\mathbb{Q}}$ of distinct real eigenvalues of $K_1^{\mathbb{Q}}$.

- A key implication of (I.3) is that, for any *MTSM* that includes M_t as pricing factors in (I.1), these macro factors must be spanned (i.e., perfectly correlated with) by $P_t^{\mathcal{N}}$:

$$M_t = \gamma_0 + \gamma_1 \, P_t^{\mathcal{N}} \, , \tag{I.4}$$

for some conformable γ_0 and γ_1 depending on W. Using (I.4), with $\gamma_1 = [\gamma_{11} \; \gamma_{12}]$, we apply the rotation:

$$Z_t = \begin{bmatrix} M_t \\ P_t^{\mathcal{L}} \end{bmatrix} = \begin{bmatrix} \gamma_0 \\ 0 \end{bmatrix} + \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ I_{\mathcal{L}} & 0_{\mathcal{L} \times (\mathcal{N}-\mathcal{L})} \end{bmatrix} P_t^{\mathcal{N}} \, , \tag{I.5}$$

to the canonical form in terms of $P_t^{\mathcal{N}}$ to obtain an equivalent model in which the factors are M_t, $P_t^{\mathcal{L}}$, r_t and Z_t (under $\mathbb{Q}$) satisfies (I.1). The specification is completed by:

$$\Delta Z_t = K_0^{\mathbb{P}} + K_1^{\mathbb{P}} \, Z_{t-1} + \sqrt{\Sigma} \, \varepsilon_t^{\mathbb{P}} \, , \quad \varepsilon_t^{\mathbb{P}} \sim N(0, I) \, . \tag{I.6}$$

I.2 – A CANONICAL GATSM

☐ **Summarizing, in the canonical form**:

- the first $\mathcal{M}$ components of Z_t are the macro variables M_t;
- risk factors are rotated so that the remaining $\mathcal{L}$ components of Z_t are the *yield portfolios* $\mathcal{P}_t^{\mathcal{L}}$ (the first $\mathcal{L}$ components of $\mathcal{P}_t^{\mathcal{N}}$);
- $r_t = \rho_0 + \rho_{1M}' M_t + \rho_{1P}' P_t^{\mathcal{L}} = \rho_0 + \rho_1' Z_t$;
- M_t is related to $\mathcal{P}_t$ through (I.4);
- Z_t follows under $\mathbb{Q}$ and $\mathbb{P}$ the Gaussian VAR(1) processes (I.1) and (I.6).

☐ **Moreover, for given W**: $\mathbb{Q}$-parameters $\left(\rho_0, \rho_1, K_0^{\mathbb{Q}}, K_1^{\mathbb{Q}}\right)$ are

explicit functions of $\Theta_{TS}^{\mathbb{Q}} = \left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, \gamma_0, \gamma_1, \Sigma\right)$.

I.2 – A CANONICAL GATSM

☐ **The canonical form reveals the** ***essential difference*** between GATSMs based entirely on $\mathcal{P}_t^{\mathcal{N}}$ and those that *includes* M_t.

- A *MSTM* with pricing factors $Z_t = (M_t, \mathcal{P}_t^{\mathcal{L}})$ offers more flexibility in fitting the joint distribution of bond yields than a pure latent factor model ($Z_t = \mathcal{P}_t^{\mathcal{N}}$).
- Indeed, the *rotation problem* of the risk factors Z_t is more severe in the latter settings. In the JSZ canonical form with pricing factors $Z_t = \mathcal{P}_t^{\mathcal{N}}$, the underlying parameter set is $\left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, K_0^{\mathcal{P}}, K_1^{\mathcal{P}}, \Sigma\right)$.
- The *MSTM* adds the spanning property (I.4) with its $\mathcal{M}$ $(\mathcal{N}+1)$ free parameters. In other words, any canonical $\mathcal{N}$-factor *MSTM* with macro factors M_t gains $\mathcal{M}$ $(\mathcal{N}+1)$ free parameters relative to pure latent factor GATSMs.
- This added flexibility is gained at a cost: M_t is linked to $\mathcal{P}_t^{\mathcal{N}}$ by (I.4).

I.3 – OBSERVATIONAL EQUIVALENCE

- Let us show that *each MTSM* where:

$$r_t = \rho_0^{\mathcal{L}} + \rho_1^{\mathcal{L}'} Z_t^{\mathcal{L}}, \tag{I.7}$$

with risk factors $Z_t^{\mathcal{L}} = (M_t', L_t')'$ (L_t being a vector of latent factors) following the Gaussian VAR(1) processes under $\mathbb{Q}$ and $\mathbb{P}$:

$$\Delta Z_t^{\mathcal{L}} = \kappa_0^{\mathbb{Q}} + \kappa_1^{\mathbb{Q}} Z_{t-1}^{\mathcal{L}} + \sqrt{\Omega}\, \varepsilon_t^{\mathbb{Q}}, \tag{I.8}$$

$$\Delta Z_t^{\mathcal{L}} = \kappa_0^{\mathbb{P}} + \kappa_1^{\mathbb{P}} Z_{t-1}^{\mathcal{L}} + \sqrt{\Omega}\, \varepsilon_t^{\mathbb{P}}, \tag{I.9}$$

is observationally equivalent to a *unique* member of *MTSM* in which $Z_t = (M_t', \mathcal{P}_t^{\mathcal{L}'})'$ with $\mathcal{L}$ yield portfolios $\mathcal{P}_t^{\mathcal{L}}$:

$$r_t = \rho_0 + \rho_1' Z_t, \tag{I.10}$$

$$\Delta Z_t = K_0^{\mathbb{Q}} + K_1^{\mathbb{Q}} Z_{t-1} + \sqrt{\Sigma}\, \varepsilon_t^{\mathbb{Q}}, \tag{I.11}$$

$$\Delta Z_t = K_0^{\mathbb{P}} + K_1^{\mathbb{P}} Z_{t-1} + \sqrt{\Sigma}\, \varepsilon_t^{\mathbb{P}}, \tag{I.12}$$

where $\left(\rho_0, \rho_1, K_0^{\mathbb{Q}}, K_1^{\mathbb{Q}}\right)$ are explicit functions of some underlying parameter set $\Theta_{TS}^{\mathbb{Q}} = \left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, \gamma_0, \gamma_1, \Sigma\right)$.

- We will make precise the sense in which $\Theta_Z = \left(\Theta_{TS}^{\mathbb{Q}}, K_0^{\mathbb{P}}, K_1^{\mathbb{P}}\right)$ uniquely characterizes the latter *MTSM*.

I.3 – OBSERVATIONAL EQUIVALENCE

- Let us assume for ease of exposition that $\kappa_1^{\mathbb{Q}}$ has nonzero, real and distinct eigenvalues with decomposition $\kappa_1^{\mathbb{Q}} = A^{\mathbb{Q}}\, diag(\lambda^{\mathbb{Q}})\, A^{\mathbb{Q}-1}$.
- JSZ adopt the following rotation:

$$\begin{aligned} X_t &= \mathcal{V}^{-1}\left[Z_t^{\mathcal{L}} + \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\right], \text{ where} \\ \mathcal{V} &= A^{\mathbb{Q}}\, diag\left[\rho_1^{\mathcal{L}\prime} A^{\mathbb{Q}}\right]^{-1}. \end{aligned} \tag{I.13}$$

- In this case we have $Z_t^{\mathcal{L}} = \mathcal{V} X_t - \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}$, thus implying:

$$\begin{aligned} r_t &= \rho_0^{\mathcal{L}} + \rho_1^{\mathcal{L}\prime} Z_t^{\mathcal{L}} \\ &= \rho_0^{\mathcal{L}} + \rho_1^{\mathcal{L}\prime} \mathcal{V} X_t - \rho_1^{\mathcal{L}\prime} \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}} \\ &= \left[\rho_0^{\mathcal{L}} - \rho_1^{\mathcal{L}\prime} \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\right] + \left(\rho_1^{\mathcal{L}\prime} A^{\mathbb{Q}}\right) diag\left[\rho_1^{\mathcal{L}\prime} A^{\mathbb{Q}}\right]^{-1} X_t \\ &= r_\infty^{\mathbb{Q}} + \mathbf{1}' X_t\,, \end{aligned} \tag{I.14}$$

where $r_\infty^{\mathbb{Q}} = \rho_0^{\mathcal{L}} - \rho_1^{\mathcal{L}\prime} \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}$.

I.3 – OBSERVATIONAL EQUIVALENCE

- Given that $\Delta Z_t^{\mathcal{L}} = \mathcal{V}\,\Delta X_t$, we have from (I.8):

$$
\begin{aligned}
\mathcal{V}\Delta X_t &= \kappa_0^{\mathbb{Q}} + \kappa_1^{\mathbb{Q}}\left[\mathcal{V}\,X_{t-1} - \left(\kappa_1^{\mathbb{Q}}\right)^{-1}\kappa_0^{\mathbb{Q}}\right] + \sqrt{\Omega}\varepsilon_t^{\mathbb{Q}}\,\text{, implying} \\
\Delta X_t &= \mathcal{V}^{-1}\,\kappa_1^{\mathbb{Q}}\,\mathcal{V}\,X_{t-1} + \mathcal{V}^{-1}\sqrt{\Omega}\varepsilon_t^{\mathbb{Q}} \\
&= diag\left[\rho_1^{\mathcal{L}\prime}\,A^{\mathbb{Q}}\right]\left[A^{\mathbb{Q}-1}\,\kappa_1^{\mathbb{Q}}\,A^{\mathbb{Q}}\right] diag\left[\rho_1^{\mathcal{L}\prime}\,A^{\mathbb{Q}}\right]^{-1} X_{t-1} \\
&\qquad + \mathcal{V}^{-1}\sqrt{\Omega}\varepsilon_t^{\mathbb{Q}} \\
&= diag\left(\lambda^{\mathbb{Q}}\right)\,X_{t-1} + \sqrt{\Sigma_X}\,\varepsilon_t^{\mathbb{Q}}\,,
\end{aligned}
\tag{I.15}
$$

where $\Omega = \mathcal{V}\sqrt{\Sigma_X}\,\mathcal{V}'$, that is $\sqrt{\Sigma_X} = \left(\mathcal{V}^{-1}\right)\,\Omega\,\left(\mathcal{V}^{-1}\right)'$.

- From relations (I.14) and (I.15), we have that the J-dimensional vector of yields Y_t is affine in X_t:

$$
Y_t = A_X\left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, \Sigma_X\right) + B_X\left(\lambda^{\mathbb{Q}}\right)\,X_t\,, \tag{I.16}
$$

with A_X and B_X obtained from standard recursions.

I.3 – OBSERVATIONAL EQUIVALENCE

- We fix a full-rank loadings matrix $W \in \mathbb{R}^{J \times J}$ such that $\mathcal{P}_t = W\, Y_t$. Focusing on the first $\mathcal{N}$ portfolios $\mathcal{P}_t^{\mathcal{N}} = W^{\mathcal{N}}\, Y_t$, we have:

$$\mathcal{P}_t^{\mathcal{N}} = W^{\mathcal{N}}\, A_X\left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, \Sigma_X\right) + W^{\mathcal{N}}\, B_X\left(\lambda^{\mathbb{Q}}\right)\, X_t\,. \tag{I.17}$$

- From (I.13) and (I.17) we have:

$$\begin{aligned}\mathcal{P}_t^{\mathcal{N}} &= W^{\mathcal{N}}\, A_X + W^{\mathcal{N}}\, B_X\, \mathcal{V}^{-1}\left[Z_t^{\mathcal{L}} + \left(\kappa_1^{\mathbb{Q}}\right)^{-1}\, \kappa_0^{\mathbb{Q}}\right] \\ &= \left[W^{\mathcal{N}}\, A_X + W^{\mathcal{N}}\, B_X\, diag\left(\rho_1^{\mathcal{L}\prime}\, A^{\mathbb{Q}}\right)\, diag\left(\lambda^{\mathbb{Q}}\right)^{-1}\, \left(A^{\mathbb{Q}}\right)^{-1}\, \kappa_0^{\mathbb{Q}}\right] \\ &\qquad + W^{\mathcal{N}}\, B_X\, \mathcal{V}^{-1}\, Z_t^{\mathcal{L}}\,,\end{aligned} \tag{I.18}$$

thus implying:

$$\begin{aligned}&\mathcal{V}\,\left(W^{\mathcal{N}}\, B_X\right)^{-1}\, \mathcal{P}_t^{\mathcal{N}} - \mathcal{V}\,\left(W^{\mathcal{N}}\, B_X\right)^{-1}\, \Big[W^{\mathcal{N}}\, A_X + \\ &W^{\mathcal{N}}\, B_X\, diag\left(\rho_1^{\mathcal{L}\prime}\, A^{\mathbb{Q}}\right)\, diag\left(\lambda^{\mathbb{Q}}\right)^{-1}\, \left(A^{\mathbb{Q}}\right)^{-1}\, \kappa_0^{\mathbb{Q}}\Big] = Z_t^{\mathcal{L}}\,.\end{aligned} \tag{I.19}$$

- We denote $\mathcal{V}_{\mathcal{M}}$ and $A_{\mathcal{M}}^{\mathbb{Q}}$ the first $\mathcal{M}$ rows of $\mathcal{V}$ and $A^{\mathbb{Q}}$, respectively. $\mathcal{V}_{\mathcal{L}}$ and $A_{\mathcal{L}}^{\mathbb{Q}}$ are the last $\mathcal{L}$ rows of $\mathcal{V}$ and $A^{\mathbb{Q}}$.

I.3 – OBSERVATIONAL EQUIVALENCE

- If we consider the first $\mathcal{M}$ rows of (I.19) we have:

$$\mathcal{V}_{\mathcal{M}} \left(W^{\mathcal{N}} B_X\right)^{-1} \mathcal{P}_t^{\mathcal{N}} - \mathcal{V}_{\mathcal{M}} \left(W^{\mathcal{N}} B_X\right)^{-1} \Big[W^{\mathcal{N}} A_X + W^{\mathcal{N}} B_X \, diag\left(\rho_1^{\mathcal{L}\prime} A^{\mathbb{Q}}\right) diag\left(\lambda^{\mathbb{Q}}\right)^{-1} \left(A^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\Big] = M_t \,, \tag{I.20}$$

and denoting $\gamma_1 = \mathcal{V}_{\mathcal{M}} \left(W^{\mathcal{N}} B_X\right)^{-1}$, we can write:

$$\begin{aligned} M_t &= \gamma_1 \, \mathcal{P}_t^{\mathcal{N}} \\ &\quad + \left[-\gamma_1 \left(W^{\mathcal{N}} A_X\right) - \mathcal{V}_{\mathcal{M}} \, diag\left[\rho_1^{\mathcal{L}\prime} A^{\mathbb{Q}}\right] diag\left(\lambda^{\mathbb{Q}}\right)^{-1} \left(A^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\right] \\ &= \left[-\gamma_1 \left(W^{\mathcal{N}} A_X\right) - A_{\mathcal{M}}^{\mathbb{Q}} \, diag\left(\lambda^{\mathbb{Q}}\right)^{-1} \left(A^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\right] + \gamma_1 \, \mathcal{P}_t^{\mathcal{N}} \\ &= \gamma_0 + \gamma_1 \, \mathcal{P}_t^{\mathcal{N}} \,, \end{aligned} \tag{I.21}$$

where:

$$\begin{aligned} \gamma_0 &= \left[-\gamma_1 \left(W^{\mathcal{N}} A_X\right) - A_{\mathcal{M}}^{\mathbb{Q}} \, diag\left(\lambda^{\mathbb{Q}}\right)^{-1} \left(A^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}}\right] \\ \gamma_1 &= \mathcal{V}_{\mathcal{M}} \left(W^{\mathcal{N}} B_X\right)^{-1} . \end{aligned} \tag{I.22}$$

I.3 – OBSERVATIONAL EQUIVALENCE

- This result allows us to write:

$$\begin{aligned} Z_t &= \begin{bmatrix} M_t \\ P_t^{\mathcal{L}} \end{bmatrix} = \Gamma_0 + \Gamma_1 \, \mathcal{P}_t^{\mathcal{N}} \\ &= \Gamma_0 + \Gamma_1 \left(W^{\mathcal{N}} A_X + W^{\mathcal{N}} B_X \, X_t \right) \\ &= \mathcal{U}_0 + \mathcal{U}_1^{-1} X_t \end{aligned} \tag{I.23}$$

where:

$$\begin{aligned} \Gamma_0 &= \begin{bmatrix} \gamma_0 \\ 0_{\mathcal{L}} \end{bmatrix}, \quad \Gamma_1 = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ I_{\mathcal{L}} & 0_{\mathcal{L} \times (\mathcal{M})} \end{bmatrix}, \\ \mathcal{U}_0 &= \Gamma_0 + \Gamma_1 \, W^{\mathcal{N}} A_X \,, \;\; \mathcal{U}_1 = \left(\Gamma_1 \, W^{\mathcal{N}} B_X \right)^{-1}, \end{aligned} \tag{I.24}$$

thus implying $X_t = \mathcal{U}_1 \, Z_t - \mathcal{U}_1 \, \mathcal{U}_0$. We thus obtain:

$$\begin{aligned} r_t &= r_\infty^{\mathbb{Q}} + \mathbf{1}' X_t = r_\infty^{\mathbb{Q}} + \mathbf{1}' \mathcal{U}_1 \, Z_t - \mathbf{1}' \mathcal{U}_1 \, \mathcal{U}_0 \,, \\ &= \left[r_\infty^{\mathbb{Q}} - \left(\mathcal{U}_1' \mathbf{1} \right)' \mathcal{U}_0 \right] + \left(\mathcal{U}_1' \mathbf{1} \right)' Z_t \\ &= \left(r_\infty^{\mathbb{Q}} - \rho_1' \mathcal{U}_0 \right) + \rho_1' Z_t = \rho_0 + \rho_1' \, Z_t \,. \end{aligned} \tag{I.25}$$

I.3 – OBSERVATIONAL EQUIVALENCE

- From (I.15), and given $\Delta X_t = \mathcal{U}_1 \, \Delta Z_t$, we also have:

$$
\begin{aligned}
\mathcal{U}_1 \, \Delta Z_t &= diag\left(\lambda^{\mathbb{Q}}\right) \mathcal{U}_1 \, (Z_{t-1} - \mathcal{U}_0) + \sqrt{\Sigma_X} \, \varepsilon_t^{\mathbb{Q}} \,, \text{ that is} \\
\Delta Z_t &= \mathcal{U}_1^{-1} \, diag\left(\lambda^{\mathbb{Q}}\right) \mathcal{U}_1 \, Z_{t-1} - \mathcal{U}_1^{-1} \, diag\left(\lambda^{\mathbb{Q}}\right) \mathcal{U}_1 \, \mathcal{U}_0 \\
&\quad + \mathcal{U}_1^{-1} \sqrt{\Sigma_X} \, \varepsilon_t^{\mathbb{Q}} \,, \\
&= K_0^{\mathbb{Q}} + K_1^{\mathbb{Q}} \, Z_{t-1} + \sqrt{\Sigma} \, \varepsilon_t^{\mathbb{Q}}
\end{aligned} \tag{I.26}
$$

where:

$$
\begin{aligned}
K_1^{\mathbb{Q}} &= \mathcal{U}_1^{-1} \, diag\left(\lambda^{\mathbb{Q}}\right) \mathcal{U}_1 \,, \quad K_0^{\mathbb{Q}} = -K_1^{\mathbb{Q}} \, \mathcal{U}_0 \,, \\
\Sigma_X &= \mathcal{U}_1 \, \Sigma \, \mathcal{U}_1' \,.
\end{aligned} \tag{I.27}
$$

- Given:

$$
\begin{aligned}
X_t &= \mathcal{V}^{-1} \left[Z_t^{\mathcal{L}} + \left(\kappa_1^{\mathbb{Q}}\right)^{-1} \kappa_0^{\mathbb{Q}} \right] \,, \text{ where} \\
Z_t &= \Gamma_0 + \Gamma_1 \, \mathcal{P}_t^{\mathcal{N}} = \mathcal{U}_0 + \mathcal{U}_1^{-1} \, X_t \,,
\end{aligned}
$$

we determine a linear mapping between Z_t and $Z_t^{\mathcal{L}}$, and it follows that the $\mathbb{P}$-dynamics of Z_t is given by (I.12).

I.3 – OBSERVATIONAL EQUIVALENCE

- Finally, from the following yield curve formula:

$$\mathcal{P}_t^{\mathcal{N}} = W^{\mathcal{N}} A_X + W^{\mathcal{N}} B_X X_t \, .$$

we obtain :

$$X_t = \left(W^{\mathcal{N}} B_X\right)^{-1} \left[\mathcal{P}_t^{\mathcal{N}} - W^{\mathcal{N}} A_X\right] \, , \tag{I.28}$$

thus implying :

$$\begin{aligned} Y_t &= A_X + B_X \left(W^{\mathcal{N}} B_X\right)^{-1} \left[\mathcal{P}_t^{\mathcal{N}} - W^{\mathcal{N}} A_X\right] \\ &= \left[I_J - B_X \left(W^{\mathcal{N}} B_X\right)^{-1} W^{\mathcal{N}}\right] A_X + B_X \left(W^{\mathcal{N}} B_X\right)^{-1} \mathcal{P}_t^{\mathcal{N}} \end{aligned} \tag{I.29}$$

- From (I.29), we observe that $W^{\mathcal{N}} Y_t \equiv \mathcal{P}_t^{\mathcal{N}}$ (ICC).

- For a formal proof of the uniqueness, see Joslin, Lee and Singleton (2013, JFE).

I.3 – OBSERVATIONAL EQUIVALENCE

☐ **To summarize**:

a) The *MTSM* with mixed macro-latent risk factors $Z_t^{\mathcal{L}}$, given by (I.7)-(I.9), is *observationally equivalent* to another *MTSM* with mixed macro-observable risk factors Z_t (yield portfolios) described by (I.10)-(I.12).

b) The primitive parameter set is $\Theta_Z = \left(\Theta_{TS}^{\mathbb{Q}}, K_0^{\mathbb{P}}, K_1^{\mathbb{P}}\right)$, with $\Theta_{TS}^{\mathbb{Q}} = \left(r_\infty^{\mathbb{Q}}, \lambda^{\mathbb{Q}}, \gamma_0, \gamma_1, \Sigma\right)$.

c) The mapping between $\left(\rho_0, \rho_1, K_0^{\mathbb{Q}}, K_1^{\mathbb{Q}}\right)$ and $\Theta_{TS}^{\mathbb{Q}}$ are:

$$\begin{aligned} \rho_1 &= \mathcal{U}_1'\mathbf{1}\,, \;\; r_\infty^{\mathbb{Q}} - \rho_1'\mathcal{U}_0\,, \\ K_1^{\mathbb{Q}} &= \mathcal{U}_1^{-1}\, diag\left(\lambda^{\mathbb{Q}}\right) \mathcal{U}_1\,, \;\; K_0^{\mathbb{Q}} = -K_1^{\mathbb{Q}}\,\mathcal{U}_0\,, \end{aligned} \tag{I.30}$$

where:

$$\begin{aligned} \mathcal{U}_0 &= \Gamma_0 + \Gamma_1\, W^{\mathcal{N}} A_X\,, \;\; \mathcal{U}_1 = \left(\Gamma_1\, W^{\mathcal{N}} B_X\right)^{-1}\,, \\ \Gamma_0 &= \begin{bmatrix} \gamma_0 \\ 0_{\mathcal{L}} \end{bmatrix}\,, \;\; \Gamma_1 = \begin{bmatrix} \gamma_{11} & \gamma_{12} \\ I_{\mathcal{L}} & 0_{\mathcal{L}\times(\mathcal{M})} \end{bmatrix}\,. \end{aligned} \tag{I.31}$$

I.4 – UNSPANNED MACRO VARIABLES

- We have seen in the previous sections that a key implication of (I.3) is that, for any *MTSM* that includes M_t as pricing factors in (I.1), these macro factors must be *spanned* (i.e., perfectly correlated with) by $P_t^{\mathcal{N}}$:

$$M_t = \gamma_0 + \gamma_1 P_t^{\mathcal{N}} , \qquad \text{(I.32)}$$

  for some conformable γ_0 and γ_1 depending on W. This relation implies that all the relevant information about the economy is captured by the current yield curve.

- The empirical evidence suggest this is not the case given that, when M_t is regressed on the first 3 PCs, the R^2 is low instead of being close to one (see Joslin, Priebsch and Singleton (2014, JF), Bauer and Rudebusch (2017, RoF)).

- In order to solve this *spanning puzzle*, JPS have proposed the following model specification:

$$\begin{aligned} r_t &= \rho_0 + 0'_{\mathcal{M}} M_t + \rho'_{1P} P_t^{\mathcal{L}} = \rho_0 + \rho'_{1P} P_t^{\mathcal{L}} , \\ \Delta P_t^{\mathcal{L}} &= K_{0,P}^{\mathbb{Q}} + K_{1,P}^{\mathbb{Q}} P_{t-1}^{\mathcal{L}} + \sqrt{\Sigma_P}\, \varepsilon_{t,P}^{\mathbb{Q}} , \quad \varepsilon_{t,P}^{\mathbb{Q}} \sim N(0, I_{\mathcal{L}}) . \end{aligned} \qquad \text{(I.33)}$$

I.4 – UNSPANNED MACRO VARIABLES

- That is, macro factors do not affect the $\mathbb{Q}$-expectation of future yield factors:
$$\mathbb{E}^{\mathbb{Q}}\left[P^{\mathcal{L}}_{t+h} \,|\, Z_t\right] = \mathbb{E}^{\mathbb{Q}}\left[P^{\mathcal{L}}_{t+h} \,|\, P^{\mathcal{L}}_{t}\right] . \tag{I.34}$$

- A consequence of (I.33) is that yields depends on yield factors only:
$$Y_t = A + B_P \, P^{\mathcal{L}}_t . \tag{I.35}$$

- Nevertheless, the $\mathbb{P}$-dynamics of Z_t is the same as spanned *MTSMs*.

Chapter II

SWITCHING REGIMES GAUSSIAN AFFINE TERM STRUCTURE MODELS

– Agenda –

II.1 – SWITCHING REGIMES CAR PROCESSES

i) Switching Regimes Univariate Car(1) Processes

- The construction is based on the following steps:

a) Let us consider the previously introduced J-states homogeneous Markov Chain (z_t). We have seen that (z_t) is an affine (Car(1)) process.

b) Let us also consider a univariate Car(1) process with a conditional Laplace transform given by $\exp\left[a(u)\, w_t + b(u)\right]$, and let us assume that $b(u)$ can be written:

$$
\begin{aligned}
b(u) &= \tilde{b}(u)'\lambda \quad \text{where} \\
\tilde{b}(u) &= (b_1(u), \ldots, b_m(u))' \text{ and } \lambda = (\lambda_1, \ldots, \lambda_m)' .
\end{aligned}
$$

c) Let us assume that the parameters λ_i are stochastic and linear functions of z_{t+1}:

$$
\lambda_i(z_{t+1}) = \lambda_i' z_{t+1} \ \forall\, i \in \{1, \ldots, m\} ,
$$

$$
\lambda_i \in \mathbb{R}^J .
$$

II.1 – SWITCHING REGIMES CAR PROCESSES

i) Switching Regimes Univariate Car(1) Processes

- Then the conditional distribution of w_{t+1}, given $\underline{w_t}$ and $\underline{z_{t+1}}$, has a Laplace transform given by:

$$\mathbb{E}[\exp(uw_{t+1}) | \, \underline{w_t} \, , \underline{z_{t+1}}] = \exp\left[a(u)\, w_t + \tilde{b}(u)'\Lambda z_{t+1}\right] \;,$$

$$\Lambda = \begin{bmatrix} \lambda_1' \\ \vdots \\ \lambda_m' \end{bmatrix} \text{ is a } (m, J)\text{-matrix.} \qquad \text{(II.1)}$$

- The conditional Laplace transform of $(w_{t+1}, z_{t+1}')'$ given $\underline{w_t} \, , \underline{z_t}$ has the following form:

$$\begin{aligned} \varphi_t(u, v) &= \mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1}) \,|\, \underline{z_t}, \; \underline{w_t}\,\right] \\ &= \exp\left\{a(u)\, w_t + a_z(v + \Lambda'\tilde{b}(u), \pi)'z_t\right\} \;, \end{aligned}$$

and thus $(w_{t+1}, z_{t+1}')'$ is a Car(1) process.

- **Proof:**

$$
\begin{aligned}
\varphi_t(u,v) &= \mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,, \underline{z_t}] \\
&= \mathbb{E}\left\{\mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,, \underline{z_{t+1}}] \mid \underline{w_t}\,, \underline{z_t}\right\} \\
&= \mathbb{E}\left\{\exp(v'z_{t+1})\,\mathbb{E}[\exp(uw_{t+1}) \mid \underline{w_t}\,, \underline{z_{t+1}}] \mid \underline{w_t}\,, \underline{z_t}\right\} \\
&= \mathbb{E}[\exp(v'z_{t+1} + a(u)\,w_t + \tilde{b}(u)'\Lambda z_{t+1}) \mid \underline{w_t}\,, \underline{z_t}] \\
&= \exp[a(u)\,w_t]\,\mathbb{E}\{\exp[(v + \Lambda'\tilde{b}(u))'z_{t+1}] \mid \underline{w_t}\,, \underline{z_t}] \\
&= \exp\left[a(u)\,w_t + a_z(v + \Lambda'\tilde{b}(u), \pi)'z_t\right]\,.
\end{aligned}
$$

Example II.1 (Switching regimes Gaussian $AR(1)$)

- *Notation:* $a(u) = u\,\rho$, $\tilde{b}(u)' = \left(u, \frac{u^2}{2}\right)$ *and* $\lambda' = (\nu, \sigma^2)$. *Thus,* $m = 2$.

- $\Lambda = \begin{bmatrix} \nu' \\ \sigma^{2\prime} \end{bmatrix}$ *is a* $(2, J)$*-matrix, where* $\nu = (\nu_1, \ldots, \nu_J)'$ *and* $\sigma^2 = (\sigma_1^2, \ldots, \sigma_J^2)'$.
- *The dynamics of* w_{t+1}*, conditionally to* $\underline{w_t}$, $\underline{z_{t+1}}$*, is given by:*

$$w_{t+1} = \nu' z_{t+1} + \rho\, w_t + (\sigma^{2\prime}\, z_{t+1})^{1/2} \varepsilon_{t+1}\,,$$

where ε_{t+1} *is a Gaussian white noise distributed as* $\mathcal{N}(0, 1)$.
- *The joint conditional Laplace transform is:*

$$\begin{aligned} \varphi_t(u, v) &= \exp\left[a(u)\, w_t + a_z(v + \Lambda' \tilde{b}(u), \pi)' z_t\right] \\ &= \exp\left[u\,\rho\, w_t + a_z\left(v + u\,\nu + \frac{u^2}{2}\,\sigma^2, \pi\right)' z_t\right] \end{aligned}$$

II.1 – SWITCHING REGIMES CAR PROCESSES

ii) Switching Regimes Univariate Car(p) Processes

- We follow the same strategy as in the previous section, but now, at the second step, we consider a univariate Index-Car(p) process with a conditional Laplace transform given by $\exp\left[a(u)\beta' W_t + b(u)\right]$.
- We assume that $b(u)$ can be written:

$$\begin{aligned} b(u) &= \tilde{b}(u)'\lambda \quad \text{where} \\ \tilde{b}(u) &= (b_1(u), \ldots, b_m(u))' \text{ and } \lambda = (\lambda_1, \ldots, \lambda_m)' . \end{aligned} \tag{II.2}$$

- At the third step, we assume that the parameters λ_i are stochastic and linear functions of $Z_t = (z_t', \ldots, z_{t-p}')'$. More precisely, we assume that the conditional distribution of w_{t+1} given $\underline{w_t}$ and $\underline{z_{t+1}}$ has a Laplace transform given by:

$$\mathbb{E}[\exp(uw_{t+1}) | \underline{w_t}, \underline{z_{t+1}}] = \exp\left[a(u)\beta' W_t + \tilde{b}(u)' \Lambda Z_t\right] ,$$

where Λ is a $[m, (p+1)J]$ matrix.

II.1 – SWITCHING REGIMES CAR PROCESSES

ii) Switching Regimes Univariate Car(p) Processes

- We assume no instantaneous causality between w_{t+1} and z_{t+1}.
- This assumption is useful at the pricing level in order to obtain simple (explicit) pricing formulas if we consider an exponential-affine SDF [Dai, Singleton and Yang (2007), Monfort and Pegoraro (2007)].
- The conditional Laplace transform of $(w_{t+1}, z'_{t+1})'$ given $\underline{w_t}$, $\underline{z_t}$ has the following form:

$$\begin{aligned} & \mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1}) \,|\, \underline{z_t},\, \underline{w_t}\,\right] \\ = \quad & \exp\left\{a(u)\beta' W_t + \left[e_1' \otimes a_z(v,\pi)' + \tilde{b}(u)'\Lambda\right] Z_t\right\} \,, \end{aligned}$$

where e_1 is the first component of the canonical basis in $\mathbb{R}^{p+1}$, and where $\otimes$ denotes the Kronecker product. The process $(w_{t+1}, z'_{t+1})'$ is thus Car($p+1$).

II.1 – SWITCHING REGIMES CAR PROCESSES

Example II.2 (Switching regimes Gaussian $AR(p)$)

- *Let us start from the AR(p) model. Its conditional Laplace transform is given by:*

$$E\left[\exp(uw_{t+1})\,|\,\underline{w_t}\right] = \exp\left[u\varphi' W_t + u\nu + \frac{\sigma^2}{2}u^2\right],$$

and the function $b(u)$ has the form (II.2) with $\tilde{b}(u)' = \left(u, \frac{u^2}{2}\right)$ and $\lambda' = (\nu, \sigma^2)$.

- *If λ is replaced by ΛZ_t, the joint process $(w_{t+1}, z'_{t+1})'$ is Car$(p+1)$ with a conditional Laplace transform given by:*

$$\begin{aligned} &\mathbb{E}\left[\exp(uw_{t+1} + v'z_{t+1})\,|\,\underline{z_t},\,\underline{w_t}\right] \\ = \quad &\exp\left[u\varphi' W_t + \left(u, \frac{u^2}{2}\right)\Lambda Z_t + a_z(v,\pi)z_t\right]. \end{aligned}$$

II.1 – SWITCHING REGIMES CAR PROCESSES

Example II.2 (Switching regimes Gaussian $AR(p)$)

- *More precisely, the dynamics is given by [using the notation $\Lambda = \begin{pmatrix} \lambda_1' \\ \lambda_2' \end{pmatrix}$]:*

$$w_{t+1} = \lambda_1' Z_t + \varphi' W_t + (\lambda_2' Z_t)^{1/2} \varepsilon_{t+1} \, ,$$

where ε_{t+1} is a Gaussian white noise distributed as $\mathcal{N}(0, \sigma^2)$, $Z_t = (z_t', \ldots, z_{t-p}')'$ and z_t is a homogeneous Markov chain.

- *In particular, let us consider the case:*

$$\Lambda = \begin{bmatrix} (1, -\varphi_1, \ldots, -\varphi_p) \otimes \nu^{*'} \\ e_1' \otimes \sigma^{*2'} \end{bmatrix}$$

and $\nu^{*'} = (\nu_1^*, \ldots, \nu_J^*)$, $\sigma^{*2'} = (\sigma_1^{*2}, \ldots, \sigma_J^{*2})$; the cond. distribution of w_{t+1} given $\underline{w_t}$ and $\underline{z_{t+1}}$ is the one corresponding to the switching AR(p) model defined by:

$$\begin{aligned} w_{t+1} - \nu^{*'} z_t \quad = \quad & \varphi_1 (w_t - \nu^{*'} z_{t-1}) + \ldots + \varphi_p (w_{t+1-p} - \nu^{*'} z_{t-p}) \\ & + (\sigma^{*'} z_t) \varepsilon_{t+1} \, . \end{aligned}$$

II.1 – SWITCHING REGIMES CAR PROCESSES

Example II.3 (Switching regimes $ARG(p)$)

- *Let us now start from the ARG(p) process associated with the conditional Laplace transform :*

$$\mathbb{E}\left[\exp(uw_{t+1})\,|\,\underline{w_t}\right] = \exp\left[\tfrac{u}{1-u\mu}\varphi' W_t - \nu\log(1-u\mu)\right] .$$

- *Here we have* $\tilde{b}(u) = -\log(1-u\mu)$ *and* $\lambda = \nu$. *If* ν *is replaced by* ΛZ_t, *where* $\Lambda Z_t > 0$, *the process* w_t *has, conditionally to the process* z_t, *a semi-strong AR(p) representation given by:*

$$w_{t+1} = \mu\Lambda Z_t + \varphi_1\, w_t\; + \;\ldots\; + \;\varphi_p\, w_{t+1-p}\; + \;\zeta_{t+1}\,,$$

where ζ_{t+1} *is a conditionally heteroscedastic martingale difference. For instance, we can take :*

$$\Lambda = e_1' \otimes \frac{\tilde{\nu}\,'}{\mu}\;,$$

where $\tilde{\nu}\,' = (\tilde{\nu}_1, \ldots, \tilde{\nu}_J)$, $\tilde{\nu}_j \geq 0$. *We have* $\Lambda Z_t = \frac{\tilde{\nu}\,'}{\mu} z_t$ *and, condit. to the process* z_t, *the process* w_t *has a semi-strong AR(p) representation given by:*

$$w_{t+1} = \tilde{\nu}\,' z_t + \varphi_1\, w_t\; + \;\ldots\; + \;\varphi_p\, w_{t+1-p}\; + \;\zeta_{t+1}\,.$$

Example II.3 (Switching regimes $ARG(p)$)

- *It is also possible to consider a Λ of the form:*

$$(1, -\varphi_1, \ldots, -\varphi_p) \otimes \frac{\tilde{\nu}'}{\mu}$$

if $\min(\tilde{\nu}_i) > \max(\tilde{\nu}_i) \sum_{i=1}^{J} \varphi_j$.

- *Indeed, in this case we have:*

$$\Lambda Z_t = \frac{1}{\mu}\left(\tilde{\nu}' z_t - \sum_{i=1}^{J} \varphi_j \tilde{\nu}' z_{t-i}\right) \geq 0$$

.

- *The weak conditional AR(p) representation is then given by:*

$$\begin{aligned} w_{t+1} - \tilde{\nu}' z_t \quad &= \quad \varphi_1 \left(w_t - \tilde{\nu}' z_{t-1}\right) + \ldots + \varphi_p \left(w_{t+1-p} - \tilde{\nu}' z_{t-p}\right) \\ &\quad + \zeta_{t+1} \,. \end{aligned}$$

II.1 – SWITCHING REGIMES CAR PROCESSES

iii) Switching Regimes Multivariate Car(p) Processes

- Switching regimes can be introduced in a multivariate Index - Car(p) model using a method extending the one retained in the univariate case.

- Let us assume that the functions $b_1(u_1)$, $b_2(u_2)$ can be written, respectively, as $\widetilde{b_1}(u_1)'\lambda_1$ and $\widetilde{b_2}(u_2)'\lambda_2$.

- Let us replace λ_1 and λ_2, respectively by $\Lambda_1 Z_t$ and $\Lambda_2 Z_t$. Then, the Laplace transform of $(w_{1,t+1}, w_{2,t+1}, z_{t+1})$, condit. to $(\underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t})$ is such that the process is Car($p+1$).

- Indeed, we obtain the following conditional Laplace transform of $(w_{1,t+1}, w_{2,t+1}, z_{t+1})$ given $(\underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t})$:

$$
\begin{aligned}
& \mathbb{E}[\exp(u_1 w_{1,t+1} + u_2 w_{2,t+1} + v' z_{t+1}) \mid \underline{w_{1,t}}, \underline{w_{2,t}}, \underline{z_t}] \\
= \quad & \exp \Big\{ [a_1(u_1)\beta_{11} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{21}]' W_{1t} \\
& \quad +[a_1(u_1)\beta_{12} + a_2(u_2 + a_1(u_1)\beta_o)\beta_{22}]' W_{2t} \\
& \quad +[e_1' \otimes a_z(v, \pi)' + \widetilde{b_1}(u_1)'\Lambda_1 + \widetilde{b_2}(u_2 + a_1(u_1)\beta_o)'\Lambda_2] Z_t \Big\} \ ,
\end{aligned}
\tag{II.3}
$$

Example II.4 (Regime Switching Gaussian $VAR(p)$ process)

- *Let us take:*

$$
\begin{aligned}
a_1(u_1) &= \rho_1 u_1, \; b_1(u_1) = \nu_1 u_1 + \frac{\sigma_1^2}{2} u_1^2, \; \tilde{b_1}(u_1)' = \left(u_1, \frac{u_1^2}{2}\right), \\
a_2(u_2) &= \rho_2 u_2, \; b_2(u_2) = \nu_2 u_2 + \frac{\sigma_2^2}{2} u_2^2, \; \tilde{b_2}(u_2)' = \left(u_2, \frac{u_2^2}{2}\right), \\
\Lambda_1 &= \begin{pmatrix} \lambda_{11}' \\ \lambda_{12}' \end{pmatrix}, \; \Lambda_2 = \begin{pmatrix} \lambda_{21}' \\ \lambda_{22}' \end{pmatrix},
\end{aligned}
$$

and let us adopt the notations $\varphi_o = \rho_1 \beta_o$, $\varphi_{11} = \rho_1 \beta_{11}$, $\varphi_{12} = \rho_1 \beta_{12}$, $\varphi_{21} = \rho_2 \beta_{21}$, $\varphi_{22} = \rho_2 \beta_{22}$.

- *We obtain the following Regime Switching Gaussian VAR(p) model:*

$$
\begin{cases}
w_{1,t+1} = \lambda_{11}' Z_t + \varphi_o w_{2,t+1} + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t} + (\lambda_{12}' Z_t)^{1/2} \eta_{1,t+1} \\
w_{2,t+1} = \lambda_{21}' Z_t + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t} + (\lambda_{22}' Z_t)^{1/2} \eta_{2,t+1} \, ,
\end{cases} \quad \text{(II.4)}
$$

where $\eta_t = (\eta_{1,t}, \eta_{2,t})'$ *is a Gaussian white noise distributed as* $\mathcal{N}(0, I_2)$, $Z_t = (z_t', \dots, z_{t-p}')'$, *and where* z_t *is a homogeneous J-state Markov chain.*

II.1 – SWITCHING REGIMES CAR PROCESSES

Example II.5 (Regime Switching $VARG(p)$ process)

- *If we take:*

$$a_1(u_1) = \frac{\rho_1 u_1}{1-u_1\mu_1}, \quad b_1(u_1) = -\nu_1 \log(1-u_1\mu_1), \quad \tilde{b_1}(u_1) = \log(1-u_1\mu_1),$$

$$a_2(u_2) = \frac{\rho_2 u_2}{1-u_2\mu_2}, \quad b_2(u_2) = -\nu_2 \log(1-u_2\mu_2), \quad \tilde{b_2}(u_2) = \log(1-u_2\mu_2),$$

we obtain the (positive) Regime Switching VARG(p) model

$$\begin{cases} w_{1,t+1} = \mu_1 \Lambda_1' Z_t + \varphi_o w_{2,t+1} + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t} + \xi_{1,t+1} \\ w_{2,t+1} = \mu_2 \Lambda_2' Z_t + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t} + \xi_{2,t+1}, \end{cases} \tag{II.5}$$

where $\xi_{1,t}$ and $\xi_{2,t}$ are non correlated, conditionally heteroscedastic, martingale differences, the conditional variances being:

$$\begin{aligned} \mathbb{V}ar[\xi_{1,t+1} \mid \underline{w_t}] &= \Lambda_1' Z_t \mu_1^2 \\ &\quad + 2\mu_1[\varphi_o(\Lambda_2' Z_t \mu_2 + \varphi_{21}' W_{1t} + \varphi_{22}' W_{2t}) + \varphi_{11}' W_{1t} + \varphi_{12}' W_{2t}] \\ \mathbb{V}ar[\xi_{2,t+1} \mid \underline{w_t}] &= \Lambda_2' Z_t \mu_2^2 + 2\mu_2(\varphi_{21}' W_{1t} + \varphi_{22}' W_{2t}). \end{aligned} \tag{II.6}$$

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs
$\mathbb{P}$-Dynamics

- The $\mathbb{P}$-dynamics of the scalar variable (w_t) is given by :

$$w_{t+1} \quad = \quad (\nu' z_t) + \varphi\, w_t + (\sigma' z_t)\varepsilon_{t+1}\,, \tag{II.7}$$

- where ε_{t+1} is a Gaussian white noise with $\mathcal{N}(0,1)$ distribution, and z_t is a J-state homogeneous Markov chain such that:

$$\mathbb{P}(z_{t+1} = e_j \mid z_t = e_i; \underline{w_t}) = \pi(e_i, e_j),$$

  e_i being the i^{th} column of the identity matrix I_J; ν and σ are J-dimensional vector of parameters.

- The absence of instantaneous causality of (z_t) to (w_t) is made to have a regime switching Gaussian AR(1) process also under the $\mathbb{Q}$ measure obtained via an exponential-affine SDF.

- In case of $\nu(z_{t+1}, z_t)$, we would need an exponential-quadratic SDF (see Gourieroux, Monfort, Pegoraro and Renne (2014)).

- Under the $\mathbb{P}$ measure, (w_{t+1}, z_{t+1}) is an affine process (see next slide).

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

$\mathbb{P}$-Dynamics

- The conditional Laplace transform of (z_t) is:

$$\varphi(v) = \mathbb{E}[\exp(v'z_{t+1})|\, z_t] = \exp\left[a_z(v,\pi)'z_t\right] ,$$

where

$$a_z(v,\pi) = \begin{bmatrix} \log\left(\sum_{j=1}^{J} \exp(v'e_j)\pi(e_1,e_j) \right) \\ \vdots \\ \log\left(\sum_{j=1}^{J} \exp(v'e_j)\pi(e_J,e_j) \right) \end{bmatrix} .$$

- The historical conditional Laplace transform of (w_{t+1}, z_{t+1}) is:

$$\begin{aligned}
\varphi_t(u,v) &= \mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,, \underline{z_t}] \\
&= \mathbb{E}\left\{\mathbb{E}[\exp(uw_{t+1} + v'z_{t+1}) \mid \underline{w_t}\,, \underline{z_{t+1}}] \mid \underline{w_t}\,, \underline{z_t}\right\} \\
&= \mathbb{E}\left\{\exp(v'z_{t+1})\,\mathbb{E}[\exp(uw_{t+1}) \mid \underline{w_t}\,, \underline{z_{t+1}}] \mid \underline{w_t}\,, \underline{z_t}\right\} \\
&= \mathbb{E}_t\left\{\exp(v'z_{t+1})\,\mathbb{E}_t\left[\exp[u((\nu'z_t) + \varphi\, w_t + (\sigma'z_t)\varepsilon_{t+1})] \mid z_{t+1}\right]\right\} \\
&= \mathbb{E}_t\left\{\exp(v'z_{t+1} + u\,(\nu'z_t) + u\,\varphi\, w_t + \tfrac{1}{2}\, u^2\,(\sigma^{2\prime}\, z_t))\right\} \\
&= \exp\left\{u\,\varphi\, w_t + \left[\left(u\,\nu + \tfrac{1}{2}\, u^2\,\sigma^2\right) + a_z(v,\pi)\right]'z_t\right\} .
\end{aligned}$$

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

The Stochastic Discount Factor

- We assume that:

$$M_{t,t+1} = \exp\left[-c\, w_t - d' z_t + \Gamma(z_t, w_t)\, \varepsilon_{t+1} - \tfrac{1}{2}\Gamma(z_t, w_t)^2 - \delta(z_t, w_t)' z_{t+1}\right] , \quad \text{(II.8)}$$

where $\Gamma(z_t, w_t) = (\gamma' z_t) + (\tilde{\gamma}' z_t)\, w_t$ and $\delta(z_t, w_t) = [\,\delta_1(z_t, w_t), \ldots, \delta_J(z_t, w_t)]'$.

- Using the absence of arbitrage assumption for the short-term interest rate between t and $t+1$, denoted by r_t and known at t, we get:

$$\begin{aligned} \exp(-r_t) &= \mathbb{E}_t\left(M_{t,t+1}\right) \\ &= \exp\left[-c\, w_t - d' z_t\right] \times \sum\nolimits_{j=1}^{J} \pi\left(e_i\,, e_j\right) \exp\left[-\delta(z_t, w_t)' e_j\right] , \end{aligned}$$

and assuming the normalization condition:

$$\sum\nolimits_{j=1}^{J} \pi\left(e_i, e_j\right) \exp\left[-\delta(z_t, w_t)' e_j\right] = 1 \quad \forall z_t, w_t\,, \quad \text{(II.9)}$$

we obtain:

$$r_t = c\, w_t + d' z_t\,. \quad \text{(II.10)}$$

- Observe that, if $\delta(z_t, w_t) = 0$ (regime shift risk is not priced), relation (II.9) is automatically satisfied.

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics

- The risk-neutral conditional Laplace transform of (w_{t+1}, z_{t+1}) is:

$$
\begin{aligned}
& \varphi_t^{\mathbb{Q}}(u,v) \\
= \quad & \mathbb{E}^{\mathbb{Q}}[\exp(uw_{t+1} + v' z_{t+1}) \mid \underline{w_t}\,,\, \underline{z_t}] \\
= \quad & \mathbb{E}_t \left[\exp\left(\Gamma(z_t, w_t)\,\varepsilon_{t+1} - \tfrac{1}{2}\Gamma(z_t, w_t)^2 - \delta(z_t, w_t)' z_{t+1}\right)\ \exp\left(u\, w_{t+1} + v' z_{t+1}\right)\right] \\
= \quad & \mathbb{E}_t \Big\{\exp\left[(v - \delta(z_t, w_t))' z_{t+1} - \tfrac{1}{2}\Gamma(z_t, w_t)^2 + u\,(\nu' z_t) + u\,\varphi\, w_t\right] \\
& \qquad \mathbb{E}_t \left[\exp\left((\Gamma(z_t, w_t) + u\,(\sigma' z_t))\,\varepsilon_{t+1}\right) \mid z_{t+1}\right]\Big\} \\
= \quad & \mathbb{E}_t \Big\{\exp\Big[(v - \delta(z_t, w_t))' z_{t+1} + u\,(\nu' z_t) + \tfrac{1}{2}\, u^2\,({\sigma^2}'\ z_t) \\
& \qquad + u\,(\varphi\, w_t + \Gamma(z_t, w_t)\,(\sigma' z_t))\Big]\Big\} \\
= \quad & \exp\Big\{u\ \left[(\nu' z_t) + (\gamma' z_t)\,(\sigma' z_t)\right] + \tfrac{1}{2}\, u^2\,({\sigma^2}'\ z_t) \\
& \qquad + u\left(\varphi + (\tilde{\gamma}' z_t)\,(\sigma' z_t)\right)\ w_t\Big\}\ \sum_{j=1}^{J} \pi(z_t, e_j)\ \exp\left[(v - \delta(w_t, z_t))' e_j\right]
\end{aligned}
$$

- for any $z_t = e_i$, the risk-neutral constant term, AR coefficient and conditional variance are given by $\nu_i^* = \nu_i + \gamma_i\,\sigma_i$, $\varphi_i^* = \varphi + \tilde{\gamma}_i\,\sigma_i$ and $\sigma_i^{2*} = \sigma_i^2$, respectively.

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics

- In order to have $\varphi_t^{\mathbb{Q}}(u,v)$ exponential-affine in (w_t, z_t), we need to impose the following conditions:

a) we must have a constant AR coefficient $\varphi^* = \varphi + \tilde{\gamma}_i \sigma_i$, that is we have to impose for any $i = 1, \ldots, J$:

$$\tilde{\gamma}_i = \frac{\varphi^* - \varphi}{\sigma_i} .$$

b) the regime indicator function z_t has to be an homogeneous Markov chain (also) under the $\mathbb{Q}$ measure, that is:

$$\mathbb{Q}(z_{t+1} = e_j \,|\, z_t = e_i) = \pi^*(e_i, e_j).$$

This condition is satisfied by imposing $\pi^*(e_i, e_j) = \pi(e_i, e_j) \exp\left(-\delta_j(w_t, e_i)\right)$, that is we must have:

$$\delta_j(w_t, e_i) = \log\left[\frac{\pi(e_i, e_j)}{\pi^*(e_i, e_j)}\right] ,$$

which implies a $\delta_j(e_i)$ function of $z_t = e_i$ and $z_{t+1} = e_j$ only, for any state i and j.

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics

- Under these conditions we have:

$$
\begin{aligned}
\varphi_t^{\mathbb{Q}}(u,v) &= \exp\left\{ u\,({\nu^*}' z_t) + \tfrac{1}{2}\,u^2\,({\sigma^{*2}}'\,z_t) + u\,\varphi^*\,w_t + a_z(v,\pi^*)' z_t \right\} \\
&= \exp\left\{ u\,\varphi^*\,w_t + \left[\left(u\,\nu^* + \tfrac{1}{2}\,u^2\,\sigma^{*2} \right) + a_z(v,\pi^*) \right]' z_t \right\} ,
\end{aligned}
$$

that is (w_t, z_t) is $\mathbb{Q}$-affine. We observe the following:

i) if $\delta(z_t, w_t) = 0$, then $\pi(e_i, e_j) = \pi^*(e_i, e_j)$;

ii) if $\pi(e_i, e_j; w_t)$ (non-homogeneous Markov chain under $\mathbb{P}$), then we must have:

$$
\delta_j(w_t, e_i) = \log\left[\frac{\pi(e_i, e_j; w_t)}{\pi^*(e_i, e_j)} \right] ,
$$

that is $\delta_j(w_t, e_i)$ remains functions of w_t.

- A typical assumption about $\pi(e_i, e_j; w_t)$ is (with $J = 2$):

$$
\mathbb{P}(z_{t+1} = e_j \,|\, z_t = e_j, w_t) = \pi(e_j, e_j; w_t) = \frac{e^{a_j + b_j' w_t}}{1 + e^{a_j + b_j' w_t}} \,,\; j \in \{1, 2\} \,, \tag{II.11}
$$

giving the possibility to explain time-varying persistence of regimes (depending on the signs of b_j and the values of w_t).

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

The ZCB Pricing Formula

- The price at date t of the zero-coupon bond with residual maturity h is :

$$B(t,h) = \exp\left(c_h\, w_t + d_h'\, z_t\right)\ ,\ \text{for}\ h \geq 1\,, \tag{II.12}$$

where c_h and d_h satisfy the following recursive equations:

$$\begin{cases} c_h = -c + \varphi^* c_{h-1} \\ \\ d_h = -d\ +\ c_{h-1}\,\nu^*\ +\ \frac{1}{2}\,c_{h-1}^2\,\sigma^{*2}\ + a_z(d_{h-1},\pi^*)\,, \end{cases} \tag{II.13}$$

and where:

$$a_z(d_{h-1},\pi^*)$$

$$= \left[\log\left(\sum_{j=1}^{J}\exp(d_{h-1}'e_j)\,\pi^*(e_1,\ e_j)\right),\ldots,\log\left(\sum_{j=1}^{J}\exp(d_{h-1}'e_j)\,\pi^*(e_J,\ e_j)\right)\right]'.$$

- Remark 1 : in the recursive equation of d_h we have $a_z(d_{h-1},\pi^*)$ instead of d_{h-1}, like in the classical single-regime GATSM.
- Remark 2 : π^* enters on $a_z(d_{h-1},\pi^*)$ only.
- Remark 3 : if $\delta(w_t,z_t) = 0$, then $\pi = \pi^*$ and $a_z(d_{h-1},\pi)$ becomes a known function of d_{h-1} once π has been estimated under the $\mathbb{P}$ measure.

II.2 – A SIMPLE SCALAR SWITCHING REGIMES GATSMs

The ZCB Pricing Formula

- **Proof:**

$$
\begin{aligned}
& B(t,h) \\
= \quad & \exp\left[c_h\, w_t + d_h'\, z_t\right] \\
= \quad & \mathbb{E}^{\mathbb{Q}}\left[\exp(-r_t)\, B(t+1, h-1)\right] \\
= \quad & \exp(-c\, w_t - d'\, z_t)\, \mathbb{E}^{\mathbb{Q}}\left[\exp\left(c_{h-1}\, w_{t+1} + d_{h-1}'\, z_{t+1}\right)\right] \\
= \quad & \exp\left[-c\, w_t - d'\, z_t + c_{h-1}\, \varphi^*\, w_t + \left(c_{h-1}\, \nu^* + \frac{1}{2}\, c_{h-1}^2\, \sigma^2 + a_z(d_{h-1}, \pi^*)\right)'\, z_t\right] \\
= \quad & \exp\left[\left(-c + c_{h-1}\, \varphi^*\right)\, w_t + \left(-d + c_{h-1}\, \nu^* + \frac{1}{2}\, c_{h-1}^2\, \sigma^2 + a_z(d_{h-1}, \pi^*)\right)'\, z_t\right]
\end{aligned}
$$

and the result is proved by identification.

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

$\mathbb{P}$-Dynamics

- The $\mathbb{P}$-dynamics of the 2-dim. vector $w_t = (w_{1,t}, w_{2,t})'$ is given by :

$$w_{t+1} = \nu(z_t) + \Phi\, w_t + \Sigma(z_t)\, \varepsilon_{t+1}\,, \tag{II.14}$$

where ε_{t+1} is a bivariate Gaussian white noise with $\mathcal{N}(0, I_2)$ distribution, and z_t is a J-state homogeneous Markov chain such that:

$$\mathbb{P}(z_{t+1} = e_j \mid z_t = e_i; \underline{w_t}) = \pi(e_i, e_j),$$

e_i being the i^{th} column of the identity matrix I_J; Φ is the $(2, 2)$ matrix of AR coefficients. We also have that:

a) $\nu(z_t) = (\nu^{(z)} \cdot z_t)$ where $\nu^{(z)} = \left[\nu^{(1)}, \ldots, \nu^{(J)}\right]$ is the $(2, J)$-dimensional matrix collecting the 2-dimensional intercepts $\nu^{(i)} = (\nu^{(z)} \cdot e_i)$ for any regime $i = 1, \ldots, J$.

b) $\Omega(z_t) = \Sigma(z_t)\, \Sigma(z_t)'$;

$\Sigma(z_t) = \sum_{i=1}^{J} (z_t' e_i)\, \Sigma_i$;

$\Omega(z_t) = \sum_{i=1}^{J} (z_t' e_i)\, \Omega_i$.

- Under the $\mathbb{P}$ measure, (w_{t+1}, z_{t+1}) is an affine process (see next slide).

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

$\mathbb{P}$-Dynamics

- The historical conditional Laplace transform of (w_{t+1}, z_{t+1}) is exponential-affine in (w_t, z_t):

$$
\begin{aligned}
& \varphi_t(u, v) \\
= \quad & \mathbb{E}[\exp(u' w_{t+1} + v' z_{t+1}) \mid \underline{w_t}, \underline{z_t}] \\
= \quad & \mathbb{E}\left\{\mathbb{E}[\exp(u' w_{t+1} + v' z_{t+1}) \mid \underline{w_t}, \underline{z_{t+1}}] \mid \underline{w_t}, \underline{z_t}\right\} \\
= \quad & \mathbb{E}\left\{\exp(v' z_{t+1})\, \mathbb{E}[\exp(u' w_{t+1}) \mid \underline{w_t}, \underline{z_{t+1}}] \mid \underline{w_t}, \underline{z_t}\right\} \\
= \quad & \mathbb{E}_t\left\{\exp(v' z_{t+1})\, \mathbb{E}_t\left[\exp[u'((\nu^{(z)} \cdot z_t) + \Phi\, w_t + \Sigma(z_t)\varepsilon_{t+1})] \mid z_{t+1}\right]\right\} \\
= \quad & \mathbb{E}_t\left\{\exp(v' z_{t+1} + u'\, (\nu^{(z)} \cdot z_t) + u'\, \Phi\, w_t + \tfrac{1}{2}\, u' \left[\textstyle\sum_{i=1}^{J} (z_t'\, e_i)\, \Omega_i\right]\; u)\right\} \\
= \quad & \exp\left\{u'\, \Phi\, w_t + \left[\left(\nu^{(z)'}\; u + F(u)\right) + a_z(v, \pi)\right]'\, z_t\right\} .
\end{aligned}
$$

where $F(u) = \left[\frac{1}{2} u'\, \Omega_1\, u, \ldots, \frac{1}{2} u'\, \Omega_J\, u\right]'$ is a J-dimensional vector.

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

The Stochastic Discount Factor

- We assume that:

$$M_{t,t+1} = \exp\left[-c'\, w_t - d' z_t + \Gamma(z_t, w_t)'\, \varepsilon_{t+1} - \tfrac{1}{2}\Gamma(z_t, w_t)'\Gamma(z_t, w_t) - \delta(z_t, w_t)' z_{t+1}\right] , \tag{II.15}$$

where $\Gamma(z_t, w_t) = \gamma(z_t) + \tilde{\Gamma}(z_t)\, w_t$ and $\delta(z_t, w_t) = [\, \delta_1(z_t, w_t), \ldots, \delta_J(z_t, w_t)]'$.

- We have that:

$\gamma(z_t) = (\gamma^{(z)} \cdot z_t)$ with $\gamma^{(z)} = \left[\gamma^{(1)}, \ldots, \gamma^{(J)}\right]$;

$\tilde{\Gamma}(z_t) = \sum_{i=1}^{J} (z_t' e_i)\, \tilde{\Gamma}^{(i)}$ is a $(2,2)$ matrix.

- Using the AAO for the short-term interest rate r_t (known at t), we get:

$$\exp(-r_t) = \exp\left[-c'\, w_t - d' z_t\right] \times \sum_{j=1}^{J} \pi\,(e_i\, , e_j) \exp\left[-\delta(z_t, w_t)' e_j\right] ,$$

and assuming :

$$\sum_{j=1}^{J} \pi\,(e_i, e_j) \exp\left[-\delta(z_t, w_t)' e_j\right] = 1 \quad \forall z_t, w_t\, , \tag{II.16}$$

we obtain:

$$r_t = c'\, w_t + d' z_t\, . \tag{II.17}$$

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics

- The risk-neutral conditional Laplace transform of (w_{t+1}, z_{t+1}) is:

$$
\begin{aligned}
& \varphi_t^{\mathbb{Q}}(u,v) = \mathbb{E}^{\mathbb{Q}}[\exp(u' w_{t+1} + v' z_{t+1}) \mid \underline{w_t}\,, \underline{z_t}] \\
= \quad & \mathbb{E}_t \left[\exp\left(\Gamma(z_t, w_t)'\,\varepsilon_{t+1} - \tfrac{1}{2}\,\Gamma(z_t, w_t)'\Gamma(z_t, w_t) - \delta(z_t, w_t)' z_{t+1}\right)\right. \\
& \qquad \left. \times \exp\left(u'\, w_{t+1} + v' z_{t+1}\right)\right] \\
= \quad & \mathbb{E}_t \left\{\exp\left[(v - \delta(z_t, w_t))' z_{t+1} - \tfrac{1}{2}\Gamma(z_t, w_t)'\Gamma(z_t, w_t) + u'\,\nu(z_t) + u'\,\Phi\, w_t\right]\right. \\
& \qquad \left. \mathbb{E}_t\left[\exp\left((\Gamma(z_t, w_t) + u'\,\Sigma(z_t))\,\varepsilon_{t+1}\right) \mid z_{t+1}\right]\right\} \\
= \quad & \mathbb{E}_t \left\{\exp\left[(v - \delta(z_t, w_t))' z_{t+1} + u'\,\nu(z_t) + \tfrac{1}{2}\,u'\,\Omega(z_t) u\right.\right. \\
& \qquad \left.\left. + u'\,(\Phi\, w_t + \Sigma(z_t)\,\Gamma(z_t, w_t))\right]\right\} \\
= \quad & \exp\left\{u'\,\left[\nu(z_t) + \Sigma(z_t)\,\gamma(z_t)\right] + \tfrac{1}{2}\,u'\,\Omega(z_t)\,u\right. \\
& \qquad \left. + u'\left(\Phi + \Sigma(z_t)\,\tilde{\Gamma}(z_t)\right)\,w_t\right\} \times \sum_{j=1}^{J} \pi(z_t, e_j)\,\exp\left[(v - \delta(w_t, z_t))' e_j\right] \\
= \quad & \exp\left\{u'\,\nu^*(z_t) + \tfrac{1}{2}\,u'\,\left[\sum_{i=1}^{J}(z_t' e_i)\,\Omega_i\right]\,u + u'\Phi^*(z_t)\,w_t\right\} \\
& \qquad \times \sum_{j=1}^{J} \pi(z_t, e_j)\,\exp\left[(v - \delta(w_t, z_t))' e_j\right] .
\end{aligned}
$$

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics

- Here, we have introduced the following notations:
  - $\nu^*(z_t) = \nu(z_t) + \Sigma(z_t)\,\gamma(z_t) = \sum_{i=1}^{J}(z_t' e_i)\left(\nu^{(i)} + \Sigma_i\,\gamma^{(i)}\right)$;
  - $\Phi^*(z_t) = \Phi + \Sigma(z_t)\,\tilde{\Gamma}(z_t) = \Phi + \sum_{i=1}^{J}(z_t' e_i)\left(\Sigma_i\,\tilde{\Gamma}^{(i)}\right)$;

  and we have $\Sigma^*(z_t) = \Sigma(z_t)$.

- As in the previously presented scalar case, the risk neutral dynamics of (w_{t+1}, z_{t+1}) is not affine and constraints must be imposed to obtain explicit pricing formulas.

a) We have to impose $\Phi^*(z_t = e_i) = \Phi^*$ for any $i = 1, \ldots, J$:

$$\tilde{\Gamma}^{(i)} = (\Sigma_i)^{-1}\,(\Phi^* - \Phi)\ .$$

b) and z_t has to be an homogeneous Markov chain (also) under the $\mathbb{Q}$ measure, that is $\mathbb{Q}(z_{t+1} = e_j \,|\, z_t = e_i) = \pi^*(e_i, e_j)$. This condition is satisfied by imposing:

$$\delta_j(w_t, e_i) = \log\left[\frac{\pi(e_i, e_j)}{\pi^*(e_i, e_j)}\right]\ ,$$

which implies a $\delta_j(e_i)$ function of $z_t = e_i$ and $z_{t+1} = e_j$ only, for any state i and j.

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

$\mathbb{Q}$-Dynamics and Pricing Formulas

- Under this conditions we have a risk neutral conditional Laplace transform that becomes exponential-affine in (w_t, z_t):

$$\varphi_t^{\mathbb{Q}}(u,v) = \exp\left\{ u'\, \Phi^*\, w_t + \left[\left(\nu^{*(z)'}\, u + F(u) \right) + a_z(v, \pi^*) \right]'\, z_t \right\} ,$$

with $\nu^{*(z)} = \left[\nu^{*(1)}, \ldots, \nu^{*(J)} \right]$ and $\nu^{*(i)} = \nu^{(i)} + \Sigma_i\, \gamma^{(i)}$ for any $i = 1, \ldots, J$. The price at date t of the zero-coupon bond with residual maturity h is :

$$B(t,h) = \exp\left(C_h'\, w_t + D_h'\, z_t \right) \ , \ \text{for } h \geq 1 \, , \tag{II.18}$$

where C_h and D_h satisfy the following recursive equations:

$$\begin{cases} C_h = -c + \Phi^{*'}\, C_{h-1} \\ D_h = -d\, + \nu^{*(z)'}\, C_{h-1}\, +\, F(C_{h-1})\, + a_z(D_{h-1}, \pi^*)\, , \end{cases} \tag{II.19}$$

and where:

$$\begin{aligned} & a_z(D_{h-1}, \pi^*) \\ = \quad & \left[\log\left(\sum_{j=1}^{J} \exp(D_{h-1}' e_j)\, \pi^*(e_1,\ e_j) \right), \ldots, \log\left(\sum_{j=1}^{J} \exp(D_{h-1}' e_j)\, \pi^*(e_J,\ e_j) \right) \right]' . \end{aligned}$$

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

About instantaneous causality of z_{t+1} to w_{t+1}

- If the $\mathbb{P}$-dynamics of the 2-dim. vector $w_t = (w_{1,t}, w_{2,t})'$ is given by :

$$w_{t+1} = \nu(z_{t+1}) + \Phi\, w_t + \Sigma(z_{t+1})\, \varepsilon_{t+1}\,, \tag{II.20}$$

- We have:

a) $\nu(z_{t+1}) = (\nu^{(z)} \cdot z_{t+1})$ where $\nu^{(z)} = \left[\nu^{(1)}, \ldots, \nu^{(J)}\right]$ is the $(2, J)$-dim. matrix collecting the 2-dim. intercepts $\nu^{(j)} = (\nu^{(z)} \cdot e_j)$ for any regime $j = 1, \ldots, J$.

b) $\Omega(z_{t+1}) = \Sigma(z_{t+1})\, \Sigma(z_{t+1})'$;
$\Sigma(z_{t+1}) = \sum_{j=1}^{J} (z'_{t+1} e_j)\, \Sigma_j$;
$\Omega(z_{t+1}) = \sum_{j=1}^{J} (z'_{t+1} e_j)\, \Omega_j$.

- Under the $\mathbb{P}$ measure, (w_{t+1}, z_{t+1}) is an affine process (see next slide).

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

About instantaneous causality of z_{t+1} to w_{t+1}

- The historical conditional Laplace transform of (w_{t+1}, z_{t+1}) is exponential-affine in (w_t, z_t):

$$
\begin{aligned}
& \varphi_t(u, v) \\
= \quad & \mathbb{E}[\exp(u' w_{t+1} + v' z_{t+1}) \mid \underline{w_t}\, , \underline{z_t}] \\
= \quad & \mathbb{E}\left\{ \mathbb{E}[\exp(u' w_{t+1} + v' z_{t+1}) \mid \underline{w_t}\, , \underline{z_{t+1}}] \mid \underline{w_t}\, , \underline{z_t} \right\} \\
= \quad & \mathbb{E}\left\{ \exp(v' z_{t+1})\, \mathbb{E}[\exp(u' w_{t+1}) \mid \underline{w_t}\, , \underline{z_{t+1}}] \mid \underline{w_t}\, , \underline{z_t} \right\} \\
= \quad & \mathbb{E}_t \left\{ \mathbb{E}_t \left[\exp[u' \left((\nu^{(z)} \cdot z_{t+1}) + \Phi\, w_t + \Sigma(z_{t+1}) \varepsilon_{t+1} \right) + v' z_{t+1}] \mid z_{t+1} \right] \right\} \\
= \quad & \mathbb{E}_t \left\{ \exp\left(v' z_{t+1} + u' \, (\nu^{(z)} \cdot z_{t+1}) + u'\, \Phi\, w_t + \tfrac{1}{2}\, u' \left[\textstyle\sum_{j=1}^{J} (z'_{t+1}\, e_j)\, \Omega_j \right] \, u \right) \right\} \\
= \quad & \exp\left\{ u'\, \Phi\, w_t + a_z \left[\nu^{(z)'}\, u + F(u) + v, \pi \right]' z_t \right\} \; .
\end{aligned}
$$

where $F(u) = \left[\frac{1}{2} u' \, \Omega_1 \, u, \ldots, \frac{1}{2} u' \, \Omega_J \, u \right]'$ is a J-dimensional vector.

II.3 – A BIVARIATE SWITCHING REGIMES GATSMs

About instantaneous causality of z_{t+1} to w_{t+1}

- The price at date t of the zero-coupon bond with residual maturity h is :

$$B(t,h) = \exp\left(C_h' \, w_t + D_h' \, z_t\right) \, , \text{ for } h \geq 1 \, , \qquad \text{(II.21)}$$

where C_h and D_h satisfy the following recursive equations:

$$\left\{ \begin{array}{l} C_h = -c + \Phi^{*'} \, C_{h-1} \\ D_h = -d \, + \, a_z \left[D_{h-1} + \nu^{*(z)'} \, C_{h-1} \, + \, F(C_{h-1}) \, , \pi^* \right] \, , \end{array} \right. \qquad \text{(II.22)}$$

and where:

$$\begin{aligned} & a_z(\ell, \pi^*) \\ = \quad & \left[\log \left(\sum_{j=1}^{J} \exp(\ell' e_j) \, \pi^*(e_1, \, e_j) \right), \ldots, \log \left(\sum_{j=1}^{J} \exp(\ell' e_j) \, \pi^*(e_J, \, e_j) \right) \right]' . \end{aligned}$$

II.4 – GENERAL UNIVARIATE SR GATSMs

Motivation: Lags and Switching Regimes for Yield Dynamics

- It is well known in the literature that interest rates show an historical dynamics involving lagged values *and* switching regimes.

- See, among the others, Hamilton (1988), Garcia and Perron (1996), Gray (1996), Boudoukh, Richardson, Smith, and Whitelaw (1999), Ang and Bekaert (2002a, 2002b), Christiansen (2004), Christiansen and Lund (2005), Cochrane and Piazzesi (2005).

- Changes in the business cycle conditions or monetary policy may affect real rates and expected inflation and cause interest rates to behave quite differently in different time periods, both in terms of level and volatility.

- In addition, there is a large empirical literature on bond yields, based in general on the class of ATSMs suggesting that regime switching models describe the term structure of interest rates better than single-regime models.

- See, for example, Bansal and Zhou (2002), Evans (2003), Ang and Bekaert (2005), Dai, Singleton and Yang (2007), Ang, Bekaert and Wei (2009).

II.4 – GENERAL UNIVARIATE SR GATSMs

Motivation: Lags and Switching Regimes for Yield Dynamics

- The yield curve is driven by a univariate or multivariate factor (w_t) which depends on its p most recent lagged values (W_t, say) and for which all the coefficients depend on the present and past values of a latent J-state non-homogeneous Markov Chain (z_t) (Z_t, say) describing different regimes in the economy.

- Consequently, the joint dynamics of (w_t, z_t) is not a Car process under the historical probability, and thus allows for nonlinearities already documented by the literature [see Ait-Sahalia (1996), Stanton (1997), Ang and Bekaert (2002b)].

- The factor (w_t) can be considered as a latent variable or an observable variable: in the second case the factor is a vector of yields.

- We consider an exponential-affine SDF with time-varying and regime-dependent risk correction coefficients which are defined as functions of the present and past values of the factor (w_t) and of the regime indicator function (z_t).

- Both factor risk and regime-shift risk are priced, and this is done by taking into account not just the information at date t, that is (w_t, z_t), but a larger information given by (W_t, Z_t).

II.4 – GENERAL UNIVARIATE SR GATSMs

$\mathbb{P}$-Dynamics

- The first set of assumptions of our Univariate Switching Regimes GATSM deals with the $\mathbb{P}$-dynamics. We assume that the historical dynamics of the latent factor (w_t) is given by :

$$w_{t+1} \quad = \quad \nu(Z_t) + \varphi_1(Z_t)\, w_t + \ldots + \varphi_p(Z_t) w_{t+1-p} + \sigma(Z_t)\varepsilon_{t+1}\,, \tag{II.23}$$

- where ε_{t+1} is a Gaussian white noise with $\mathcal{N}(0,1)$ distribution,
- $Z_t = (z_t', \ldots, z_{t-p}')'$, $W_t = (w_t, \ldots, w_{t+1-p})'$,
- z_t is a J-state non-homogeneous Markov chain such that:

$$\mathbb{P}(z_{t+1} = e_j \,|\, z_t = e_i; \underline{w_t}) = \pi(e_i, e_j\,; W_t),$$

  e_i being the i^{th} column of the identity matrix I_J.

- Equation (II.23) will be also written

$$w_{t+1} \quad = \quad \nu(Z_t) + \varphi(Z_t)' W_t + \sigma(Z_t)\varepsilon_{t+1}\,, \tag{II.24}$$

where $\varphi(Z_t) = (\varphi_1(Z_t), \ldots, \varphi_p(Z_t))'$.

II.4 – GENERAL UNIVARIATE SR GATSMs

$\mathbb{P}$-Dynamics

- This model can also be rewritten in the following vectorial form:

$$W_{t+1} = \Phi(Z_t)W_t + [\nu(Z_t) + \sigma(Z_t)\varepsilon_{t+1}]\, e_1 \qquad \text{(II.25)}$$

where

$$\Phi(Z_t) = \begin{bmatrix} \varphi_1(Z_t) & \dots & \dots & \varphi_p(Z_t) \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 1 & 0 \end{bmatrix}$$

is a $(p \times p)$-matrix, and where e_1 is the first column of the identity matrix I_p.

- Note that, since the coefficients φ_i are allowed to depend on Z_t and since the Markov chain z_t may not be homogeneous, the dynamics of (w_t, z_t) is not Car in general.

- It is not Car even if we assume constant autoregressive coefficients.

The Stochastic Discount Factor

- We assume that:

$$M_{t,t+1} = \exp\left[-c'W_t - d'Z_t + \Gamma(Z_t, W_t)\,\varepsilon_{t+1} - \frac{1}{2}\Gamma(Z_t, W_t)^2 - \delta(Z_t, W_t)'z_{t+1}\right] , \quad \text{(II.26)}$$

where $\Gamma(Z_t, W_t) = \gamma(Z_t) + \tilde{\gamma}'(Z_t)W_t$ and $\delta(Z_t, W_t) = [\,\delta_1(Z_t, W_t), \dots, \delta_J(Z_t, W_t)]'$.

- Using the absence of arbitrage assumption for the short-term interest rate between t and $t+1$, denoted by r_t and known at t, we get:

$$\exp(-r_{t+1}) = \mathbb{E}_t\left(M_{t,t+1}\right)$$

$$= \exp\left[-c'W_t - d'Z_t\right] \times \sum_{j=1}^{J} \pi\left(e_i, e_j\,; W_t\right) \exp\left[-\delta(Z_t, W_t)'e_j\right] ,$$

and assuming the normalization condition:

$$\sum_{j=1}^{J} \pi\left(e_i, e_j\,; W_t\right) \exp\left[-\delta(Z_t, W_t)'e_j\right] = 1 \quad \forall Z_t, W_t , \qquad \text{(II.27)}$$

we obtain:

$$r_t = c'W_t + d'Z_t\,. \qquad \text{(II.28)}$$

II.4 – GENERAL UNIVARIATE SR GATSMs

Risk Premia

- If we denote with p_t the price of a given asset at time t, the risk premium of this asset between t and $t+1$ is:

$$\omega_t = \log(\mathbb{E}_t p_{t+1}) - \log p_t - r_{t+1}.$$

- In particular, the risk premium between t and $t+1$ of an asset providing the payoff $\exp(-\theta\, w_{t+1})$ at $t+1$ is :

$$\omega_t(\theta) = \theta\, \Gamma(Z_t, W_t)\sigma(Z_t)\,. \tag{II.29}$$

- and, if we consider a digital asset providing one money unit at $t+1$ if $z_{t+1} = e_j$, its risk premium between t and $t+1$ is given by :

$$\omega_t(\theta) = \delta_j(Z_t, W_t)\,. \tag{II.30}$$

II.4 – GENERAL UNIVARIATE SR GATSMs

$\mathbb{Q}$-Dynamics

- The risk-neutral dynamics of the process (w_t, z_t) is given by:

$$w_{t+1} \quad = \quad \nu(Z_t) + \gamma(Z_t)\sigma(Z_t) + [\varphi(Z_t) + \tilde{\gamma}(Z_t)\sigma(Z_t)]' W_t + \sigma(Z_t)\xi_{t+1} \, , \tag{II.31}$$

  where ξ_{t+1} is a Gaussian white noise with $\mathcal{N}(0,1)$ distribution, and z_t is a Markov chain such that:

$$\mathbb{Q}(z_{t+1} = e_j \,|\, \underline{z_t}; \underline{w_t}) = \pi\left(z_t, e_j \,; W_t\right) \exp\left[(-\delta(Z_t, W_t))' e_j\right] \, .$$

  Note that, from (III.7), these probabilities add to one.

- In order to get a generalized linear term structure we impose that the risk-neutral dynamics is switching regime gaussian Car(p). Using (II.2), this impose that the dynamics has to satisfy the following specification:

$$w_{t+1} \quad = \quad {\nu^*}' Z_t + {\varphi^*}' W_t + ({\sigma^*}' Z_t)\xi_{t+1} \, , \tag{II.32}$$

  where z_t is a J-state Markov chain such that:

$$\mathbb{Q}(z_{t+1} = e_j \,|\, z_t = e_i) = \pi^*\left(e_i, e_j\right) \, . \tag{II.33}$$

II.4 – GENERAL UNIVARIATE SR GATSMs
$\mathbb{Q}$-Dynamics

- This implies the following restrictions on the historical dynamics and on the SDF:

i) $\sigma(Z_t) = \sigma^{*'} Z_t$:

the historical stochastic volatility must be linear in Z_t;

ii) $\gamma(Z_t) = \dfrac{\nu^{*'} Z_t - \nu(Z_t)}{\sigma^{*'} Z_t}$:

for given historical stochastic drift $\nu(Z_t)$ and stochastic volatility $\sigma^{*'} Z_t$, the coefficient $\gamma(Z_t)$ belongs to the previous family indexed by the free parameter vector ν^*.

iii) $\tilde{\gamma}(Z_t) = \dfrac{\varphi^* - \varphi(Z_t)}{\sigma^{*'} Z_t}$:

for given historical stochastic slope parameter $\varphi(Z_t)$ and stochastic volatility $\sigma^{*'} Z_t$ the vector $\tilde{\gamma}(Z_t)$ belongs to the previous family indexed by the free parameter vector φ^*.

II.4 – GENERAL UNIVARIATE SR GATSMs

$\mathbb{Q}$-Dynamics

iv) $\delta_j(Z_t, W_t) = \log \left[\dfrac{\pi\,(z_t, e_j\,;W_t)}{\pi^*(z_t, e_j)} \right]$:

for a given historical transition matrix $\pi\,(z_t, e_j\,;W_t)$, the coefficient $\delta_j(Z_t, W_t)$ depends on z_t only and belongs to the previous family indexed by the entries $\pi^*(z_t, e_j)$ of a transition matrix.

- Note that condition *iv)* implies the normalization condition (III.7).
- We have seen in the previous section that the risk-neutral dynamics is defined by relations (II.32), (II.33); relation (II.32) can be rewritten:

$$W_{t+1} \quad = \quad \Phi^* W_t + \left[{\nu^*}' Z_t + ({\sigma^*}' Z_t)\xi_{t+1} \right] e_1 \qquad \text{(II.34)}$$

where

$$\Phi^* = \begin{bmatrix} \varphi_1^* & \dots & \dots & \varphi_p^* \\ 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & \dots & 1 & 0 \end{bmatrix} \text{ is a } (p \times p) - \text{matrix}\,,$$

with $W_t = (w_t, \dots, w_{t+1-p})'$ and where e_1 is the first column of the identity matrix I_p.

II.4 – GENERAL UNIVARIATE SR GATSMs

The Generalized Linear Term Structure

- The price at date t of the zero-coupon bond with residual maturity h is :

$$B(t,h) = \exp\left(C_h' \, W_t + D_h' \, Z_t\right) \; , \; \text{for } h \geq 1 \, , \qquad \text{(II.35)}$$

where the vectors C_h and D_h satisfy the following recursive equations :

$$\begin{cases} C_h = \Phi^{*'} \, C_{h-1} - c \\ D_h = -d \, + \, C_{1,h-1} \, \nu^* \, + \, \frac{1}{2} \, C_{1,h-1}^2 \, \sigma^{*2} \, + \tilde{D}_{h-1} + F(D_{1,h-1}) \, , \end{cases} \qquad \text{(II.36)}$$

- $C_{1,h-1}$ = first component of the p-dim. vector C_{h-1},
- $D_{h-1} = (D_{1,h-1}', D_{2,h-1}')'$, $\tilde{D}_{h-1} = (D_{2,h-1}', 0)'$,
- $D_{1,h-1}$ = the first J-dim. component of D_{h-1},
- $D_{2,h-1}$ = the remaining $(p\,J)$-dimensional component of D_{h-1};
- $F(D_{1,h-1}) = e_1 \otimes a_z(D_{1,h-1}, \, \pi^*)$, e_1 being the vector $(1, 0, \dots, 0)'$ of size $(p+1)$ and a_z:

$$\begin{aligned} & a_z(D_{1,h-1}, \pi^*) \\ = \quad & \left[\log\left(\sum_{j=1}^{J} \exp(D_{1,h-1}' e_j) \pi^*(e_1, \; e_j) \right), \dots, \log\left(\sum_{j=1}^{J} \exp(D_{1,h-1}' e_j) \pi^*(e_J, \; e_j) \right) \right]' . \end{aligned}$$

II.4 – GENERAL UNIVARIATE SR GATSMs

The Generalized Linear Term Structure

- σ^{*2} is the vector whose components are the squares of the entries of σ^*. The initial conditions are $C_0 = 0$, $D_0 = 0$ (or $C_1 = -c$, $D_1 = -d$).
- The yield-to-maturity formula is:

$$R(t,h) \quad = \quad -\frac{1}{h}\log B(t,h) = -\frac{C_h'}{h}W_t - \frac{D_h'}{h}Z_t\,, \quad h \geq 1\,. \tag{II.37}$$

- So, they are linear functions of the p-dimensional vector W_t and of the $(p+1)J$-dimensional vector Z_t. This means that, the term structure at date t depends on the present and past values of w_t and z_t, and not just on their values in t.
- Moreover, we observe that there is, in general, instantaneous causality between $R(t,h)$ and z_t.
- If $w_t = r_t$, the previous results remain valid, and the only modification comes from the absence of arbitrage opportunity condition for r_t, which imposes:

$$c = e_1\,, d = 0\,, \tag{II.38}$$

with e_1 the first column of the identity matrix I_p. Consequently, the initial conditions in the recursive equations become:

$$C_1 = -e_1\,, D_1 = 0\,. \tag{II.39}$$

II.5 – GENERAL MULTIVARIATE SR GATSMs

- For sake of notational simplicity we consider the two factor case but an extension to more that two factors is straightforward. The historical dynamics of $\tilde{w}_t = (w_{1,t}, w_{2,t})'$ is a bivariate Switching Regimes Gaussian VAR(p) model given by:

$$
\begin{aligned}
w_{1,t+1} &= \nu_1(Z_t) + \varphi_o(Z_t) w_{2,t+1} + \varphi_{11}(Z_t)' W_{1t} + \varphi_{12}(Z_t)' W_{2t} \\
&\qquad + \sigma_1(Z_t)\varepsilon_{1,t+1} \\
w_{2,t+1} &= \nu_2(Z_t) + \varphi_{21}(Z_t)' W_{1t} + \varphi_{22}(Z_t)' W_{2t} + \sigma_2(Z_t)\varepsilon_{2,t+1} \,,
\end{aligned}
\tag{II.40}
$$

- where $\varepsilon_{1,t}$ and $\varepsilon_{2,t}$ are independent standard normal white noises,
- $W_{1t} = (w_{1,t}, \ldots, w_{1,t+1-p})'$, $W_{2t} = (w_{2,t}, \ldots, w_{2,t+1-p})'$,
- $Z_t = (z_t', \ldots, z_{t-p}')'$,
- with z_t a J-state non-homogeneous Markov chain such that

$$
\mathbb{P}(z_{t+1} = e_j \,|\, z_t = e_i; \underline{\tilde{w}_t}) = \pi(e_i, e_j\,; \tilde{W}_t)
$$

- and where $\tilde{W}_t = (W_{1t}', W_{2t}')'$.

II.5 – GENERAL MULTIVARIATE SR GATSMs

- The recursive form (II.40) is equivalent to the canonical form:

$$\left\{\begin{array}{lcl} w_{1,t+1} & = & \tilde{\nu}_1(Z_t) + \tilde{\varphi}_{11}(Z_t)' W_{1t} + \tilde{\varphi}_{12}(Z_t)' W_{2t} \\ & & \qquad + \sigma_1(Z_t)\varepsilon_{1,t+1} + \varphi_o(Z_t)\sigma_2(Z_t)\varepsilon_{2,t+1} \\ w_{2,t+1} & = & \nu_2(Z_t) + \varphi_{21}(Z_t)' W_{1t} + \varphi_{22}(Z_t)' W_{2t} + \sigma_2(Z_t)\varepsilon_{2,t+1} \, , \end{array}\right. \tag{II.41}$$

where $\tilde{\nu}_1 = \nu_1 + \varphi_o \nu_2$, $\tilde{\varphi}_{11} = \varphi_{11} + \varphi_o \varphi_{21}$, $\tilde{\varphi}_{12} = \varphi_{12} + \varphi_o \varphi_{22}$, or, with obvious notations:

$$\tilde{w}_{t+1} \quad = \quad \tilde{\nu}(Z_t) + \tilde{\Phi}(Z_t)\tilde{W}_t + S(Z_t)\varepsilon_{t+1} \, , \tag{II.42}$$

where

$$S(Z_t) = \begin{bmatrix} \sigma_1(Z_t) & \varphi_o(Z_t)\sigma_2(Z_t) \\ 0 & \sigma_2(Z_t) \end{bmatrix} .$$

- Let us introduce the following the notations:

$$\Gamma(Z_t, \tilde{W}_t) = \left[\Gamma_1(Z_t, \tilde{W}_t), \, \Gamma_2(Z_t, \tilde{W}_t) \right]' \, ,$$

where $\Gamma_i(Z_t, \tilde{W}_t) = \gamma_i(Z_t) + \tilde{\gamma}_i(Z_t)' \tilde{W}_t \, , i \in \{1, 2\}$, and $\Gamma(Z_t, \tilde{W}_t) = \gamma(Z_t) + \tilde{\Gamma}(Z_t)\tilde{W}_t$,

with $\gamma(Z_t) = [\gamma_1(Z_t), \gamma_2(Z_t)]'$ and $\tilde{\Gamma}(Z_t) = [\tilde{\gamma}_1(Z_t)', \tilde{\gamma}_2(Z_t)']'$.

II.5 – GENERAL MULTIVARIATE SR GATSMs

- Then, the one-period SDF $M_{t,t+1}$ is defined as:

$$M_{t,t+1} = \exp\Big[-c'\tilde{W}_t - d'Z_t + \Gamma(Z_t, \tilde{W}_t)'\,\varepsilon_{t+1} - \tfrac{1}{2}\Gamma(Z_t, \tilde{W}_t)'\Gamma(Z_t, \tilde{W}_t) - \delta(Z_t, \tilde{W}_t)'z_{t+1}\Big]\,.$$

- Assuming the normalization condition (III.7) and the absence of arbitrage opportunity for r_t we get:

$$r_t = c'\tilde{W}_t + d'Z_t\,. \tag{II.43}$$

- It is also easily seen that the risk premium for an asset providing the payoff $\exp(-\theta'\tilde{w}_{t+1})$ at $t+1$ is $\omega(\theta) = \theta' S(Z_t)\Gamma(Z_t, \tilde{W}_t)$ and that the risk premium associated with the digital payoff $\mathbb{I}_{(e_j)}(z_{t+1})$ is unchanged.
- The risk-neutral dynamics of the process $(\tilde{w}_t, z_t)$ is given by:

$$\tilde{w}_{t+1} = \tilde{\nu}(Z_t) + S(Z_t)\gamma(Z_t) + [\tilde{\Phi}(Z_t) + S(Z_t)\tilde{\Gamma}(Z_t)]\tilde{W}_t + S(Z_t)\xi_{t+1}\,, \tag{II.44}$$

where ξ_{t+1} is (under $\mathbb{Q}$) a bivariate Gaussian white noise with $\mathcal{N}(0, I_2)$ distribution, and where $Z_t = (z_t', \ldots, z_{t-p}')'$, z_t being a Markov chain such that:

$$\mathbb{Q}(z_{t+1} = e_j \mid \underline{z_t}; \underline{\tilde{w}_t}) = \pi(z_t, e_j\,; \tilde{W}_t)\exp\Big[(-\delta(Z_t, \tilde{W}_t))'e_j\Big]\,.$$

II.5 – GENERAL MULTIVARIATE SR GATSMs

- If we want to obtain a Switching bivariate Car process in the risk-neutral world, we must have using similar arguments as in the univariate case :

i) $\sigma_1(Z_t) = {\sigma_1^*}' Z_t \,, \;\; \sigma_2(Z_t) = {\sigma_2^*}' Z_t \,, \;\; \varphi_o(Z_t) = \varphi_o^*$, and thus:

$$S(Z_t) = \begin{bmatrix} {\sigma_1^*}' Z_t & \varphi_o^* {\sigma_2^*}' Z_t \\ 0 & {\sigma_2^*}' Z_t \end{bmatrix} = S^*(Z_t)$$

ii) $\gamma(Z_t) = [S(Z_t)]^{-1}[\nu^* Z_t - \tilde{\nu}(Z_t)]$, where ν^* is a $(2 \times (p+1)\, J)$-matrix.

iii) $\tilde{\Gamma}(Z_t) = [S(Z_t)]^{-1}\left[\Phi^* - \tilde{\Phi}(Z_t)\right]$, where Φ^* is a $(2 \times 2\, p)$-matrix.

iv) $\delta_j(Z_t, \tilde{W}_t) = \log\left[\dfrac{\pi(z_t, e_j\,;\tilde{W}_t)}{\pi^*(z_t, e_j)}\right].$

II.5 – GENERAL MULTIVARIATE SR GATSMs

- The risk-neutral dynamics can be written:

$$\left\{ \begin{array}{rcl} w_{1,t+1} & = & \nu_1^* Z_t + \Phi_1^* \tilde{W}_t + S_1^*(Z_t)\xi_{t+1} \\ w_{2,t+1} & = & \nu_2^* Z_t + \Phi_2^* \tilde{W}_t + S_2^*(Z_t)\xi_{t+1} \, , \end{array} \right. \tag{II.45}$$

where ν_i^*, Φ_i^*, S_i^* are the i^{th} row of ν^*, Φ^*, S^*, with $i \in \{1,2\}$. Or,

$$\tilde{W}_{t+1} \;=\; \tilde{\Phi}^* \tilde{W}_t + \left[\nu_1^* Z_t + S_1^*(Z_t)\xi_{t+1}\right] e_1 + \left[\nu_2^* Z_t + S_2^*(Z_t)\xi_{t+1}\right] e_{p+1} \, ,$$

e_1 (respectively, e_{p+1}) is of size $2p$, with entries equal to zero except the first (respectively, the $(p+1)^{th}$) one which is equal to one, and

$$\tilde{\Phi}^* = \begin{bmatrix} \Phi_{11}^* & \Phi_{12}^* \\ \widetilde{I} & \widetilde{\mathbf{0}} \\ \Phi_{21}^* & \Phi_{22}^* \\ \widetilde{\mathbf{0}} & \widetilde{I} \end{bmatrix}$$

- $\Phi_1^* = (\Phi_{11}^*, \Phi_{12}^*)$, $\Phi_2^* = (\Phi_{21}^*, \Phi_{22}^*)$, $\widetilde{\mathbf{0}}$ is a $[(p-1) \times p]$-matrix of zeros
- and $\widetilde{I}$ is a $[(p-1) \times p]$-matrix equal to $(I_{p-1}, 0)$, where 0 is a vector of size $(p-1)$.

II.5 – GENERAL MULTIVARIATE SR GATSMs

- The price at date t of the zero-coupon bond with residual maturity h is :

$$B(t,h) = \exp\left(C_h' \, \tilde{W}_t + D_h' \, Z_t \right) \; , \; \text{for } h \geq 1 \qquad \text{(II.46)}$$

where the vectors C_h and D_h satisfy the following recursive equations:

$$\begin{cases} C_h = \tilde{\Phi}^{*'} \, C_{h-1} - c \\ D_h = -d + C_{1,h-1} \, \nu_1^{*'} + C_{p+1,h-1} \, \nu_2^{*'} + \frac{1}{2} \, C_{1,h-1}^2 (\sigma_1^{*2} + \varphi_o^{*2} \sigma_2^{*2}) \\ + (C_{1,h-1})(C_{p+1,h-1}) \varphi_o^{*2} \sigma_2^{*2} + \frac{1}{2} \, C_{p+1,h-1}^2 \sigma_2^{*2} + \tilde{D}_{h-1} + F(D_{1,h-1}) \, , \end{cases} \qquad \text{(II.47)}$$

and where $\tilde{D}_{h-1}$ and $F(D_{1,h-1})$ have the same meaning as we have seen before, and the initial conditions are $C_0 = 0$, $D_0 = 0$ (or $C_1 = -c$, $D_1 = -d$). The yields to maturity are:

$$R(t,h) \quad = \quad -\frac{C_h'}{h} \, \tilde{W}_t - \frac{D_h'}{h} \, Z_t \, , \quad h \geq 1 \, . \qquad \text{(II.48)}$$

II.5 – GENERAL MULTIVARIATE SR GATSMs

- In the case of observable factor, we can take $w_{1t} = r_t$, and $w_{2t} = R(t, H)$ for a given time to maturity H. In this case the absence of arbitrage conditions for r_t and $R(t, H)$ imply:

$$\begin{aligned} (i)\ C_1 &= -e_1,\ D_1 = 0\,,\ \text{or}\ \ c = e_1,\ d = 0 \\ (ii)\ C_H &= -H\, e_{p+1},\ D_H = 0\,. \end{aligned} \tag{II.49}$$

- Conditions (i) are used as initial values in the recursive relations (C_h, D_h), and conditions (ii) implies restrictions on the parameters $\tilde{\Phi}^*, \nu_1^*, \nu_2^*$ which must be taken into account at the estimation stage.

- Note that, the number of constrains is less than the number of additional parameters appearing in the SDF and, therefore, the historical dynamics is not further constrained.

II.6 – EMPIRICAL ANALYSIS

Observable Factor Approach

- In the present empirical analysis we consider an observable factor (yields at different maturities), because this approach has several important advantages (already seen in the previous Lecture) .

- In order to study and to precisely understand the role played by lags and switching regimes in the term structure modeling, we start the empirical analysis by estimating (single-regime) Gaussian VAR(p) Factor-Based ATSM [denoted VARN(p)], in a bivariate (short rate, and spread between the long and short rate) setting.

- The historical parameters are estimated by exact Maximum Likelihood, while the risk-neutral parameters are estimated by nonlinear least squares (NLLS) applied to the yield-to-maturity formulas.

- The second step of the empirical illustration concerns the estimation of bivariate Switching Regime Gaussian VAR(p) Factor-Based ATSM [denoted SVARN(p)], where the latent variable (z_t) is assumed to be a two-state Markov chain.

II.6 – EMPIRICAL ANALYSIS

Observable Factor Approach

- As in the single-regime specifications, the factor is given by the short rate and spread. The historical parameters are estimated by maximization of the likelihood function calculated using the Kitagawa-Hamilton filter.

- The risk-neutral parameters are estimated by Constrained NLLS applied on the yield-to-maturity formulas, after extraction of the latent variable $Z_t = (z_t, \ldots, z_{t-p})'$ using smoothed probabilities calculated using a generalization of the Kim's Smoothing algorithm [see Kim (1994) for the basic algorithm and Monfort and Pegoraro (2006b) for the generalization].

- The CRSP data set on the U. S. term structure of interest rates [treasury zero-coupon bond (ZCB) yields], that we consider in the following application, covers the period from June 1964 to December 1995 and contains 379 monthly observations for each of the nine maturities : 1, 3, 6 and 9 months and 1, 2, 3, 4 and 5 years.

II.6 – EMPIRICAL ANALYSIS
Estimated Models

- We start with the estimation of the single-regime GATSM. The historical dynamics of the factor (w_t) is given by the following Gaussian VAR(p) process :

$$\begin{aligned} w_{t+1} &= \nu + \varphi_1 \, w_t + \ldots + \varphi_p \, w_{t+1-p} + \sigma \, \varepsilon_{t+1} \\ &= \nu + \varphi W_t + \sigma \varepsilon_{t+1} \,, \end{aligned} \tag{II.50}$$

- where ε_{t+1} is an 2-dimensional Gaussian white noise with $\mathcal{N}(0, I_2)$ distribution [I_2 denotes the (2×2) identity matrix];
- σ and φ_j, for each $j \in \{1, \ldots, p\}$, are (2×2) matrices [σ can be chosen, for instance, lower triangular],
- $\varphi = [\varphi_1, \ldots, \varphi_p]$ is an $(2 \times 2p)$ matrix; ν is an 2-dimensional vector;
- $W_t = (w_t', \ldots, w_{t+1-p}')'$ is an $(2p)$-dimensional vector, and $p \in \{1, 2\}$.

II.6 – EMPIRICAL ANALYSIS
Estimated Models

- Then, we move to the regime-switching setting and we estimate the $\text{SVARN}^{nh}_{mv}(p)$ Factor-Based ATSM. The $\mathbb{P}$-dynamics of (w_t) is:

$$\begin{aligned} w_{t+1} &= \nu(z_t) + \varphi_1\, w_t + \ldots + \varphi_p\, w_{t+1-p} + \eta_{t+1} \\ &= \nu(z_t) + \varphi W_t + \mathcal{A}(z_t)'\eta_{t+1}\,, \end{aligned} \tag{II.51}$$

- where (η_t) is a bivariate Gaussian white noise with $\mathcal{N}(0, I_2)$ distribution,
- $\mathcal{A}(z_t)$ is an upper triangular matrix, (z_t) is a 2-state non-homogeneous Markov chain and $p \in \{1, 2\}$.

- We denote by $\Sigma(z_t) = \mathcal{A}(z_t)'\mathcal{A}(z_t)$ the (conditional to z_t and W_t) variance-covariance matrix of w_{t+1}.

- The SDF between the date t and $t+1$ is given by (II.26).

- The notations (nh) and (mv) in $\text{SVARN}^{nh}_{mv}(p)$ indicate that switching regimes are (under $\mathbb{P}$) described by a non-homogeneous (nh) Markov chain applying to both the conditional mean and volatility (mv) of (w_t).

II.6 – EMPIRICAL ANALYSIS

Estimated Models

- The transition probabilities have the following logistic form:

$$\mathbb{P}(z_{t+1} = e_j \mid z_t = e_j, w_t) = \pi(e_j, e_j; x_t) = \frac{e^{a_j + b_j' w_t}}{1 + e^{a_j + b_j' w_t}}, \; j \in \{1, 2\}. \tag{II.52}$$

- We will see, in the following sections, that the regime $z_t = e_1$ will be identified with a low (L) volatility regime and $z_t = e_2$ will be identified with an high (H) volatility regime of the factor (w_t).

- The state-dependence of the historical transition probabilities gives the possibility to explain the time-varying persistence of volatility regimes empirically observed in the interest rates historical dynamics [see, for instance, Ang and Bekaert (2002b) and Dai, Singleton and Yang (2006)].

- Note that, since the Markov chain is not homogeneous, the process $(w_{t+1}, z_{t+1}')'$ is not Car.

II.6 – EMPIRICAL ANALYSIS

Estimated Models

- The methodology we follow to estimate the parameters of both single-regime and switching regime models is based on a consistent two-step procedure.

- As far as the single-regime GATSM estimation is concerned, in the first step, thanks to observations on the 2-dimensional observable factor (w_t), we estimate the vector of parameters:

$$\theta_{\mathbb{P}} = [\, \nu' \, , vec(\varphi)' \, , vech(\sigma\sigma')']'$$

  characterizing the historical dynamics (w_t), by exact Maximum Likelihood (ML).

- With regard to the estimation of bivariate $\text{SVARN}^{nh}_{mv}(p)$ ATSM, in the first step, using the observations on the observable factor (w_t), the vector of historical parameters

$$\tilde{\theta}_{\mathbb{P}} = [\, \nu(e_1)', \nu(e_2)', vec(\varphi)', vech[\mathcal{A}(e_1)], vech[\mathcal{A}(e_2)], a_1, b_1', a_2, b_2']\,'$$

  is estimated by the maximization of the likelihood function calculated by means of the Kitagawa-Hamilton filter [see Hamilton (1994)].

II.6 – EMPIRICAL ANALYSIS

Estimated Models

- In the second step, using observations on yields with maturities different from those used in the first step and for a given estimates of historical parameters, we estimate the vector of risk-neutral parameters $\theta_{\mathbb{Q}}$ by (Constrained) NLLS.

- $Z_t = (z_t, \dots, z_{t-p})'$ in the yield formula of the SVARN$^{nh}_{mv}(p)$ model, is extracted using smoothed probabilities calculated with the Kim's smoothing algorithm.

- We define the short rate as the one-month yield to maturity $[r_t = R(t,1)]$. In our empirical illustration the factor is given by:

$$w_t = [R(t,1), R(t,60) - R(t,1)]' ,$$

where $[R(t,60) - R(t,1)]$ is the spread at date t between the five-years and one-month yield to maturity [see also Ang and Bekaert (2002a, 2002b), and Ang, Piazzesi and Wei (2006)].

II.6 – EMPIRICAL ANALYSIS

Estimated Models

- At the end of the first estimation step, models are ranked in terms of the Akaike Information Criterion (AIC), while, after the second estimation step, models are ranked in terms of root mean square errors (RMSE) and absolute errors.

- The constrained NLLS estimator is given by :

$$\begin{cases} \hat{\theta}_{\mathbb{Q}} = Arg \min_{\theta_{\mathbb{Q}}} S^2(\theta_{\mathbb{Q}}) \\ S^2(\theta_{\mathbb{Q}}) = \sum_{t=p}^{T} \sum_{h \in H_2^*} [\tilde{R}(t,h) - R(t,h)]^2, \\ \text{s. t. } \sum_{t=p}^{T} [\tilde{R}(t,60) - R(t,60)]^2 = 0 \,, \end{cases} \tag{II.53}$$

where $R(t,h)$ is the theoretical yield determined by formula (II.48).

- The constraint in the minimization program (II.53) guarantees the ICC on the five-year yield to maturity.

II.6 – EMPIRICAL ANALYSIS

Results: Single-regime case

- $\mathbb{P}$-dynamics of w_t : (φ_1, φ_2) significantly different from zero.
- Univariate (Ljung-Box) and joint (Portmanteau) autocorrelation analysis: serial dependence is better explained when $p = 2$, but null hypothesis rejected for univariate and joint residuals.
- $\mathbb{Q}$-dynamics of w_t: $(\varphi_1^*, \varphi_2^*)$ significantly different from zero [RMSE reduces when AR order p increases].
- Pricing error comparison: VAR(2) model provides good performances (better than 2-Factor CIR, 3-Factor CIR, 3-Factor $\mathbb{A}_1(3)$ models; as well as 2-Factor RS-CIR model of Bansal and Zhou (2002, JF)).
- Expectation Hypothesis Puzzle (EHP) over short and long horizons : when we move from VAR(1) to VAR(2) specification, model implied-$\phi_1(h, m)$ becomes negative and/or closer to empirical value but, when $h \in \{4, 5\}$ years is regressed on $m = 3$ years $\Rightarrow \phi_1(h, m) > 0$.

II.6 – EMPIRICAL ANALYSIS

Results: 2-regime case

- $\mathbb{P}$-dynamics of w_t: (φ_1, φ_2) STILL significantly different from zero.
- Univariate (Ljung-Box) and joint (Portmanteau) autocorrelation analysis: serial dependence is better explained when $p = 2$; null hypothesis is accepted for $p = 2$ and for univariate (short rate, spread) and joint residuals; determinant role of Lags and Switching Regimes.

- $\mathbb{Q}$-dynamics of w_t: $(\varphi_1^*, \varphi_2^*)$ STILL significantly different from zero [RMSE smaller w.r.t. VAR models, and it reduces when AR order p increases].
- Pricing error comparison: $\text{SVARN}_{mv}^{nh}(p)$ models perform well.

- Expectation hypothesis puzzle (EHP) over short and long horizons:

i) for each p, $\text{SVARN}_{mv}^{nh}(p)$ model performs better than VAR(p) model; important role played by Switching Regimes (in particular, over the long horizon);

ii) when we move from $\text{SVARN}_{mv}^{nh}(1)$ to $\text{SVARN}_{mv}^{nh}(2)$ specification: model implied-$\phi_1(h, m)$ becomes (more) negative and closer to empirical value and $\text{SVARN}_{mv}^{nh}(2)$ specification is always able to explain EHP.

Annualized Absolute Pricing Errors [basis points]

	2-dim CIR	2-dim RS-CIR	3-dim CIR	3-dim AF	2-dim VAR(1)	2-dim VAR(2)	2-dim $SVARN^{nh}_{mv}(1)$	2-dim $SVARN^{nh}_{mv}(2)$
Mean	30	23	25	25	25	23	23	23
Std. Dev.	18	16	21	22	17	16	16	16
Min.	3	3	1	2	3	3	3	3
Max.	121	114	133	137	108	91	100	95

II.6 – EMPIRICAL ANALYSIS

Pricing Lags and Regime-Shift Risks

- We want to study at which level pricing the risk associated
  - to the second lagged factor value,
  - or to regime shifts,

  is important in explaining interest rates dynamics.

- We compare Campbell-Shiller regr. coeff. of model SVARN$_{mv}^{nh}(2)$,
  - with those obtained if we impose $\varphi_2 = \varphi_2^*$ [SVARN$_{cm}^{nh}(2)$ model]; the risk associated to the second lagged factor value is not priced;
  - or if we consider $\pi_t(e_j, e_j; r_t) = \pi(e_j, e_j) = \pi^*(e_j, e_j)$, $\forall\, j \in \{1, 2\}$ and $\forall\, t$ [SVARN$_{mv}^{ch}(2)$ model]; regime-shift risk is not priced.

II.6 – EMPIRICAL ANALYSIS

Pricing Lags and Regime-Shift Risks

Short Horizon	$m = 3$ months	$m = 6$ months	$m = 9$ months
$h = 6$ months	-0.6942 (0.2533)		
2-Factor SVARN$^{nh}_{mv}(2)$	-0.5323 (0.3195)		
2-Factor SVARN$^{nh}_{cm}(2)$	-0.3747 (0.3107)		
2-Factor SVARN$^{ch}_{mv}(2)$	-0.1736 (0.2963)		
$h = 9$ months	-0.8863 (0.3238)	-0.4023 (0.2429)	
2-Factor SVARN$^{nh}_{mv}(2)$	-0.5957 (0.3286)	-0.4928 (0.2555)	
2-Factor SVARN$^{nh}_{cm}(2)$	-0.4446 (0.3158)	-0.4522 (0.2439)	
2-Factor SVARN$^{ch}_{mv}(2)$	-0.2472 (0.3008)	-0.2443 (0.2429)	
$h = 12$ months	-1.3226 (0.3530)	-0.7867 (0.2381)	-0.4371 (0.1312)
2-Factor SVARN$^{nh}_{mv}(2)$	-0.6591 (0.3363)	-0.5731 (0.2574)	-0.4906 (0.2297)
2-Factor SVARN$^{nh}_{cm}(2)$	-0.5113 (0.3216)	-0.5224 (0.2467)	-0.4703 (0.2226)
2-Factor SVARN$^{ch}_{mv}(2)$	-0.3170 (0.3057)	-0.3168 (0.2443)	-0.2817 (0.2239)

II.6 – EMPIRICAL ANALYSIS

Pricing Lags and Regime-Shift Risks

Long Horizon	$m = 1$ year	$m = 2$ years	$m = 3$ years
$h = 4$ years	-1.8078 (0.2981)	-0.8380 (0.2889)	-0.0421 (0.2682)
2-Factor $\text{SVARN}^{nh}_{mv}(2)$	-1.3067 (0.2857)	-0.7469 (0.2657)	-0.0306 (0.2756)
2-Factor $\text{SVARN}^{nh}_{cm}(2)$	-1.0239 (0.2886)	-0.2757 (0.2748)	**0.6756** (0.2775)
2-Factor $\text{SVARN}^{ch}_{mv}(2)$	-1.1758 (0.2799)	-0.8034 (0.2633)	-0.3788 (0.2832)
$h = 5$ years	-1.7470 (0.3291)	-0.9720 (0.3199)	-0.2378 (0.3283)
2-Factor $\text{SVARN}^{nh}_{mv}(2)$	-1.5368 (0.3208)	-0.8823 (0.2963)	-0.0472 (0.3015)
2-Factor $\text{SVARN}^{nh}_{cm}(2)$	-1.1081 (0.3196)	-0.2237 (0.2987)	**0.7983** (0.2910)
2-Factor $\text{SVARN}^{ch}_{mv}(2)$	-1.5074 (0.3198)	-1.0864 (0.2965)	-0.6075 (0.3118)

II.6 – EMPIRICAL ANALYSIS

The Expectation Hypothesis Puzzle

Short Horizon	$m = 3$ months	$m = 6$ months	$m = 9$ months
$h = 6$ months	-0.6942 (0.2533)		
2-Factor CIR	0.8253 (1.2208)		
2-Factor RS-CIR	-**0.5360** (0.5243)		
3-Factor CIR	1.1676 (1.1952)		
3-Factor AF	1.4916 (1.3874)		
2-Factor VAR(1)	0.5828 (0.3485)		
2-Factor VAR(2)	-0.3800 (0.3837)		
2-Factor SVARN$_{mv}^{nh}$(1)	-0.1339 (0.3068)		
2-Factor SVARN$_{mv}^{nh}$(2)	-**0.5323** (0.3195)		

Short Horizon	$m = 3$ months	$m = 6$ months	$m = 9$ months
$h = 9$ months	-0.8863 (0.3238)	-0.4023 (0.2429)	
2-Factor CIR	0.7367 (1.2166)	0.6982 (1.2043)	
2-Factor RS-CIR	-**0.5757** (0.5632)	-0.3021 (0.5819)	
3-Factor CIR	1.0632 (1.1903)	1.0287 (1.1809)	
3-Factor AF	1.4348 (1.3692)	1.4316 (1.3380)	
2-Factor VAR(1)	0.4133 (0.3469)	0.4722 (0.2693)	
2-Factor VAR(2)	-0.5480 (0.3960)	-**0.3890** (0.3182)	
2-Factor SVARN$^{nh}_{mv}$(1)	-0.1421 (0.3033)	0.1526 (0.2334)	
2-Factor SVARN$^{nh}_{mv}$(2)	-**0.5957** (0.3286)	-**0.4928** (0.2555)	

II.6 – EMPIRICAL ANALYSIS
The Expectation Hypothesis Puzzle

Short Horizon	$m = 3$ months	$m = 6$ months	$m = 9$ months
$h = 12$ months	-1.3226 (0.3530)	-0.7867 (0.2381)	-0.4371 (0.1312)
2-Factor CIR	0.6455 (1.2137)	0.6068 (1.2008)	0.5713 (1.1889)
2-Factor RS-CIR	-0.6000 (0.6030)	-0.3145 (0.6186)	-0.0679 (0.6630)
3-Factor CIR	0.9524 (1.1874)	0.9185 (1.1783)	0.8833 (1.1659)
3-Factor AF	1.3763 (1.3514)	1.3741 (1.3210)	1.3692 (1.2959)
2-Factor VAR(1)	0.2454 (0.3486)	0.3187 (0.2710)	0.3796 (0.2430)
2-Factor VAR(2)	**-0.6935** (0.4069)	**-0.5272** (0.3248)	**-0.3675** (0.2930)
2-Factor SVARN$_{mv}^{nh}$(1)	-0.1767 (0.3001)	0.0411 (0.2325)	0.1743 (0.2103)
2-Factor SVARN$_{mv}^{nh}$(2)	**-0.6591** (0.3363)	**-0.5731** (0.2574)	**-0.4906** (0.2297)

II.6 – EMPIRICAL ANALYSIS

The Expectation Hypothesis Puzzle

Long Horizon	$m = 1$ year	$m = 2$ years	$m = 3$ years
$h = 4$ years	-1.8078 (0.2981)	-0.8380 (0.2889)	-0.0421 (0.2682)
2-Factor CIR	-0.6928 (1.1713)	**-0.7324** (1.0755)	-0.7406 (0.9935)
2-Factor RS-CIR	-0.1848 (1.0852)	-0.0309 (1.1429)	**-0.0631** (1.1527)
3-Factor CIR	-0.8191 (1.1752)	-0.7238 (1.0336)	-0.6292 (0.9036)
3-Factor AF	0.5652 (1.1344)	0.5962 (1.0712)	0.6078 (1.0239)
2-Factor VAR(1)	-0.8569 (0.3536)	-0.0085 (0.3414)	0.8626 (0.3514)
2-Factor VAR(2)	**-1.4088** (0.4084)	-0.2338 (0.3864)	0.9368 (0.3843)
2-Factor $\text{SVARN}^{nh}_{mv}(1)$	-0.8036 (0.2952)	-0.3602 (0.3017)	0.0865 (0.3393)
2-Factor $\text{SVARN}^{nh}_{mv}(2)$	**-1.3067** (0.2857)	**-0.7469** (0.2657)	**-0.0306** (0.2756)

II.6 – EMPIRICAL ANALYSIS

The Expectation Hypothesis Puzzle

Long Horizon	$m = 1$ year	$m = 2$ years	$m = 3$ years
$h = 5$ years	-1.7470 (0.3291)	-0.9720 (0.3199)	-0.2378 (0.3283)
2-Factor CIR	-1.1314 (1.1825)	**-1.1130** (1.0514)	-1.0652 (0.9434)
2-Factor RS-CIR	-0.4527 (1.1837)	-0.3607 (1.2056)	**-0.4120** (1.1877)
3-Factor CIR	-1.4772 (1.1850)	-1.2330 (0.9957)	-1.0217 (0.8326)
3-Factor AF	0.2358 (1.1134)	0.2958 (1.0508)	0.3304 (1.0017)
2-Factor VAR(1)	-1.1444 (0.4102)	-0.0033 (0.3953)	1.1279 (0.3970)
2-Factor VAR(2)	**-1.6686** (0.4635)	-0.2112 (0.4373)	1.2060 (0.4267)
2-Factor SVARN$_{mv}^{nh}$(1)	-1.1375 (0.3463)	-0.6103 (0.3525)	**-0.0748** (0.3873)
2-Factor SVARN$_{mv}^{nh}$(2)	**-1.5368** (0.3208)	**-0.8823** (0.2963)	**-0.0472** (0.3015)

II.6 – EMPIRICAL ANALYSIS

Regime Probabilities and Business Cycle

We can interpret, in general, low (L) and high (H) volatility regimes in (r_t) as different stages of the business cycle :

a) $\mathbb{P}(z_{t+1} = e_2 = H \,|\, z_t = e_2 = H, r_t)$ increases when r_t increases.
$\mathbb{P}(z_{t+1} = e_1 = L \,|\, z_t = e_1 = L, r_t)$ decreases when r_t increases.

b) When close to the end of expansionary phase :
short rates rise faster than long rates [= negative spread] $\rightarrow$ recession;
$\hat{\pi}_t^{L\rightarrow H} = \mathbb{P}(z_{t+1} = H \,|\, z_t = L, r_t; \hat{\theta}) \uparrow$,
$\hat{\pi}_t^{H\rightarrow L} = \mathbb{P}(z_{t+1} = L \,|\, z_t = H, r_t; \hat{\theta}) \downarrow$,
$w_{2,t}$ moves down towards negative values;

c) If we are in a recessionary period :
typically short rates go down to induce expansion $\rightarrow$ positive spread
$\hat{\pi}_t^{L\rightarrow H} = \mathbb{P}(z_{t+1} = H \,|\, z_t = L, r_t; \hat{\theta}) \downarrow$,
$\hat{\pi}_t^{H\rightarrow L} = \mathbb{P}(z_{t+1} = L \,|\, z_t = H, r_t; \hat{\theta}) \uparrow$,
$w_{2,t}$ rises towards positive values;

FIGURE 4 – Persistence Probability in regime 1 (solid) and 2 (dashed);

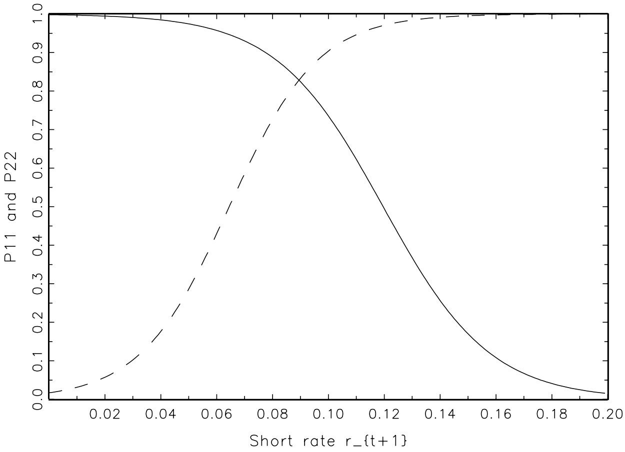


FIGURE 4a – Smoothed Probability of High Volatility Regime;
SARN$_2^{nh}$(1) and SARN$_2^{nh}$(5) models;

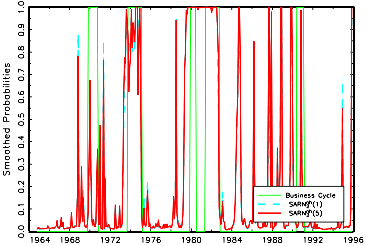


FIGURE 4b – Transition Probability
from low to high volatility regime;
SARN$_2^{nh}$(1) and SARN$_2^{nh}$(5) models;

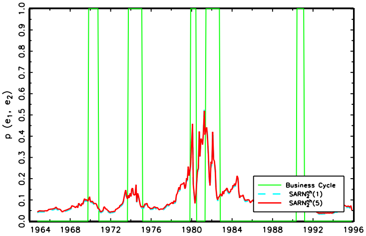


FIGURE 4c – Conditional Mean of the short rate

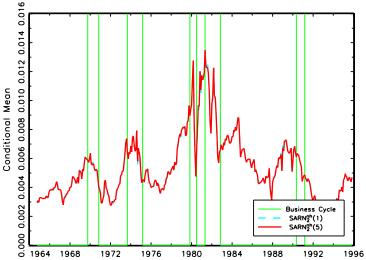


FIGURE 4d – Transition Probability
from high to low volatility regime;
SARN$_2^{nh}$(1) and SARN$_2^{nh}$(5) models;

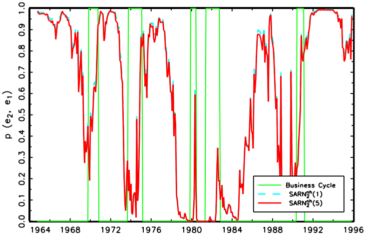


FIGURE 5a – Smoothed Probability of High Volatility Regime:
$SVARN_2^{nh}(1)$ and $SVARN_2^{nh}(2)$ models;

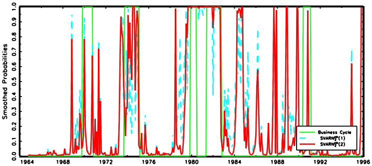


FIGURE 5b – Transition Probability
from low to high volatility regime;
$SVARN_2^{nh}(1)$ and $SVARN_2^{nh}(2)$ models;

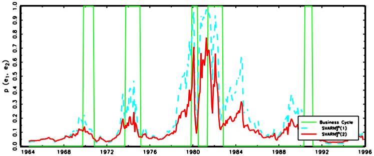


FIGURE 5c – Conditional Mean of the short rate

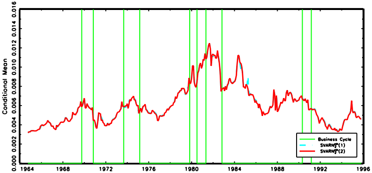


FIGURE 5d – Transition Probability
from high to low volatility regime;
$SVARN_2^{nh}(1)$ and $SVARN_2^{nh}(2)$ models;

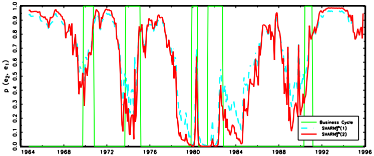
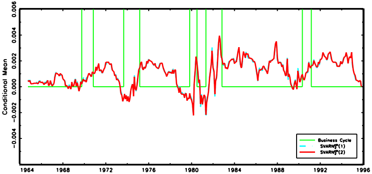


FIGURE 5e – Conditional Mean of the spread



Chapter III

NON-NEGATIVE AFFINE TERM STRUCTURE MODELS

– Agenda –

III.1 – AUTOREGRESSIVE GAMMA ATSMs

The $\mathbb{P}$-dynamics

- Single regime or multiple regimes Gaussian (VAR) ATSMs do not (theoretically) guarantee that the yield-to-maturity formula generate positive yields for any date t, residual maturity h, any parameter values and realization of w_t.

- If assume that the factor w_t follows an Autoregressive Gamma (ARG) process (with a well specified SDF), then :

  - the term structure of interest rates will be **affine** in the factor (w_t) (or W_t, if $p > 1$).

  - any model implied yield-to-maturity will be **strictly positive**.

- We consider the case of a **scalar** and **latent** factor, with $p = 1$.

- Let us assume that the scalar latent factor w_t has a dynamics, under the historical probability $\mathbb{P}$, described by an ARG(1) process.

III.1 – AUTOREGRESSIVE GAMMA ATSMs

The $\mathbb{P}$-dynamics

- Under $\mathbb{P}$, the Laplace transform of w_{t+1}, conditionally to $\underline{w_t}$, is given by:

$$\begin{aligned}\mathbb{E}\left[\exp(uw_{t+1}) \,|\, \underline{w_t}\right] &= \exp\left[\frac{\rho u}{1-u\mu}\, w_t - \nu \log(1-u\mu)\right] , \\ &= \exp\left[a(u;\rho,\mu)\, w_t + b(u;\nu,\mu)\right] .\end{aligned}$$

- and it has this semi-strong positive AR(1) form:

$$w_{t+1} = \nu\mu + \rho w_t + \varepsilon_{t+1} ,$$

where $\{\varepsilon_t\}$ is a conditionally heteroscedastic martingale difference, with $\mathbb{E}(\varepsilon_{t+1} \,|\, \varepsilon_t) = 0$ and conditional variance is $\mathbb{V}ar(\varepsilon_{t+1} \,|\, \varepsilon_t) = \nu\mu^2 + 2\mu\rho w_t$, and whose conditional Laplace transform is given by:

$$\begin{aligned}\mathbb{E}\left[\exp(u\varepsilon_{t+1}) \,|\, \underline{\varepsilon_t}\right] &= \mathbb{E}\left\{\exp\left[u(w_{t+1} - \nu\mu - \rho\, w_t) \,|\, \underline{w_t}\right]\right\} , \\ &= \exp\left[a(u;\rho,\mu)\, w_t + b(u;\nu,\mu) - u\,(\nu\,\mu + \rho\, w_t)\right] , \\ &= \exp\left[(a(u;\rho,\mu) - u\rho)\, w_t + b(u;\nu,\mu) - u\,\nu\,\mu\right] .\end{aligned}$$

III.1 – AUTOREGRESSIVE GAMMA ATSMs
The Stochastic Discount Factor

- The one-period SDF $M_{t,t+1}$ is assumed to be given by:

$$\begin{aligned} M_{t,t+1} \quad &= \quad \exp\left[-\beta - \alpha w_t + \Gamma\,\varepsilon_{t+1}\right] \\ &\qquad \times \exp\left[-a\left(\Gamma;\rho,\mu\right)\,w_t - b\left(\Gamma;\nu,\mu\right) + \Gamma\left(\nu\,\mu + \rho\,w_t\right)\right] \end{aligned}$$

  with risk-correction coefficient given by Γ.

- It is built in such a way that:
  - $\dfrac{d\,\mathbb{Q}_{t,t+1}}{d\,\mathbb{P}_{t,t+1}} = \dfrac{M_{t,t+1}}{\mathbb{E}_t[M_{t,t+1}]}$ is a density :
    $\dfrac{M_{t,t+1}}{\mathbb{E}_t[M_{t,t+1}]} > 0$ and $\mathbb{E}_t\left[\dfrac{M_{t,t+1}}{\mathbb{E}_t[M_{t,t+1}]}\right] = 1$;
  - the no-arbitrage restriction is explicitly satisfied :
    $\mathbb{E}_t[M_{t,t+1}] = \exp(-\,r_t)$ if and only if $r_t = \beta + \alpha\,w_t$.

III.1 – AUTOREGRESSIVE GAMMA ATSMs
The Stochastic Discount Factor

- Let us consider the functions:

$$a(u;\rho,\mu) = \frac{\rho u}{1-u\mu} \quad \text{and} \quad b(u;\nu,\mu) = -\nu\log(1-u\mu)\,;$$

  then, we have:

- **Lemma :**

$$a(u+g;\rho,\mu) - a(g;\rho,\mu) = a(u;\rho^*,\mu^*)$$

$$b(u+g;\nu,\mu) - b(g;\nu,\mu) = b(u;\nu,\mu^*)$$

$$\text{with } \rho^* = \frac{\rho}{(1-g\mu)^2},\ \mu^* = \frac{\mu}{1-g\mu}\,,$$

  [Proof : exercise] and we will consider the case $g = \Gamma$.

III.1 – AUTOREGRESSIVE GAMMA ATSMs

The Affine Positive Term Structure of Interest Rates

- The price at date t of the zero-coupon bond with time to maturity h is :

$$B(t,h) = \exp(c_h\, w_t + d_h)\,, \quad h \geq 1\,,$$

where c_h and d_h satisfies, for $h \geq 1$, the recursive equations:

$$\begin{cases} c_h &= -\alpha + [a(c_{h-1} + \Gamma; \rho, \mu) - a(\Gamma; \rho, \mu)] \\ &= -\alpha + a(c_{h-1}; \rho^*, \mu^*)\,, \\ d_h &= -\beta + [b(c_{h-1} + \Gamma; \nu, \mu) - b(\Gamma; \nu, \mu)] + d_{h-1} \\ &= -\beta + b(c_{h-1}; \nu, \mu^*) + d_{h-1}\,, \end{cases}$$

with initial conditions $c_0 = 0, d_0 = 0$ (or $c_1 = -\alpha, d_1 = -\beta$). If $w_t = r_t$, then $c_1 = -1$ and $d_1 = 0$.

- The affine term structure of interest rates is:

$$R(t,h) = -\frac{1}{h}\log B(t,h) = -\frac{c_h}{h}\, w_t - \frac{d_h}{h}\,, \quad h \geq 1\,.$$

III.1 – AUTOREGRESSIVE GAMMA ATSMs

The Positive $\mathbb{Q}$-dynamics

- The risk-neutral Laplace transform of w_{t+1}, conditionally to $\underline{w_t}$, is:

$$\begin{aligned} \mathbb{E}_t^{\mathbb{Q}}[\exp(uw_{t+1})] &= \mathbb{E}_t\left[\frac{M_{t,t+1}}{\mathbb{E}_t(M_{t,t+1})}\exp(uw_{t+1})\right] \\ &= \exp\left\{[a(u+\Gamma;\rho,\mu)-a(\Gamma;\rho,\mu)]\ w_t+[b(u+\Gamma;\nu,\mu)-b(\Gamma;\nu,\mu)]\right\} \\ &= \exp\left[a(u;\rho^*,\mu^*)\,w_t+b(u;\nu,\mu^*)\right] \end{aligned}$$

- Under $\mathbb{Q}$, w_{t+1} is the following positive semi-strong AR(1) process:

$$w_{t+1} \quad = \quad \nu\,\mu^* + \rho^* w_t \ + \eta_{t+1}\,,$$

with $\rho^* = \dfrac{\rho}{(1-\Gamma\,\mu)^2} > 0$ and $\mu^* = \dfrac{\mu}{1-\Gamma\,\mu} > 0$, and where η_{t+1} is such that $\mathbb{E}(\eta_{t+1}\,|\,\eta_t) = 0$ and $\mathbb{V}ar(\eta_{t+1}\,|\,\eta_t) = \nu(\mu^*)^2 + 2\mu^*\rho^* w_t$.

III.1 – AUTOREGRESSIVE GAMMA ATSMs

The Positive $\mathbb{Q}$-dynamics

- **Positivity of the yields :**

- Since $r_t = R(t,1) = \beta + \alpha\, w_t$, and since w_t is a positive process, the short rate process will be positive as soon as β and α are nonnegative.

- The positivity of r_t implies that of $R(t,h)$, at any date t and time to maturity h, because $R(t,h) = -\frac{1}{h}\log E_t^{\mathbb{Q}}[\exp(-r_t - \dots \; -r_{t+h-1})]$.

- This is the discrete-time equivalent of the (continuous-time affine) Cox-Ingersoll-Ross (1985) model.

III.2 – SWITCHING REGIMES ARG ATSMs

Modelling strategy

- In this section we start building our class of Switching Regimes ARG ATSMs following the Risk-Neutral Constrained Direct Modelling strategy.

- First, we assume that $y_t = (w_t, z_t')'$ (say) follows, under $\mathbb{P}$, a general Switching Autoregressive Gamma of order p ($SwARG(p)$) process. More precisely:
  - (w_t) is a quantitative factor and its $\mathbb{P}$-dynamics, conditionally to $\underline{y_{t-1}}$, is given by an $ARG(p)$ process.
  - (z_t) is, under $\mathbb{P}$, a J-state non-homogeneous Markov Chain taking values $e_j \in \mathbb{R}^J$, $j \in \{1, \ldots, J\}$, where e_j is the j^{th} column of the $(J \times J)$ identity matrix.
  - The process $(w_t, z_t')'$ is not Car under $\mathbb{P}$.

- Second, we specify an exponential-affine Stochastic Discount Factor (SDF) $M_{t,t+1}$, with time-varying and regime-dependent risk correction coefficients which are function of the present and past values of $(w_t, z_t')'$.

III.2 – SWITCHING REGIMES ARG ATSMs

Modelling strategy

- Third, we impose to the $\mathbb{Q}$-dynamics of $(w_t, z_t')'$ to be a Switching Autoregressive Gamma Car(p) process in order to obtain an explicit yield-to-maturity formula.

- This specification is obtained by imposing restrictions to historical and SDF parameters in such a way that the $\mathbb{Q}$-dynamics be Car(p).

- Roughly speaking:
  - we start by specifying a general non-linear (non-Car) $\mathbb{P}$-dynamics,
  - we then assume that its Car counterpart be the $\mathbb{Q}$-dynamics and, finally,
  - we impose a set of restrictions on historical and risk-neutral parameters in such a way that the exponential-affine SDF be able to (explicitly) guarantee the matching between them.

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{P}$-dynamics

- We assume that the Laplace transform of the conditional distribution of w_{t+1}, given $(\underline{w_t}, \underline{z_t})$, is:

$$\mathbb{E}\left[\exp(u w_{t+1}) \,|\, \underline{w_t}, , \underline{z_t}\right] = \exp\left[\frac{u}{1-u\mu(W_t,Z_t)}\left[\varphi_1(Z_t)\, w_t + \ldots + \varphi_p(Z_t) w_{t-p+1}\right]\right.$$
$$\left. - \nu(Z_t)\log(1-u\mu(W_t,Z_t))\right] , \tag{III.1}$$

where $Z_t = (z_t', \ldots, z_{t-p}')'$, $W_t = (w_t, \ldots, w_{t+1-p})'$, and where z_t is a J-states non-homogeneous Markov chain such that:

$$\mathbb{P}(z_{t+1} = e_j \,|\, z_t = e_i; \underline{w_t}) = \pi(e_i, e_j\, ; W_t).$$

- Using the notation:

$$\begin{aligned} A[u; \varphi(Z_t), \mu(W_t, Z_t)] &= \frac{u}{1-u\mu(W_t,Z_t)}[\varphi_1(Z_t), \ldots, \varphi_p(Z_t)]' \\ &= \frac{u}{1-u\mu(W_t,Z_t)}\varphi(Z_t) \\ b[u; \nu(Z_t), \mu(W_t, Z_t)] &= -\nu(Z_t)\log(1-u\mu(W_t,Z_t)) , \end{aligned}$$

relation (III.1) can be written:

$$\mathbb{E}\left[\exp(u w_{t+1}) \,|\, \underline{w_t}, , \underline{z_t}\right] = \exp\left\{A[u; \varphi(Z_t), \mu(W_t,Z_t)]' W_t + b[u; \nu(Z_t), \mu(W_t,Z_t)]\right\} . \tag{III.2}$$

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{P}$-dynamics

- The process (w_t) can also be written:

$$
\begin{aligned}
w_{t+1} &= \nu(Z_t)\,\mu(X_t, Z_t) + \varphi_1(Z_t)\,w_t + \ldots + \varphi_p(Z_t)\,w_{t+1-p} + \varepsilon_{t+1} \\
&= \nu(Z_t)\mu(X_t, Z_t) + \varphi(Z_t)' W_t + \varepsilon_{t+1}\,,
\end{aligned}
\tag{III.3}
$$

where ε_{t+1} is a martingale difference sequence with conditional Laplace transform given by:

$$
\begin{aligned}
&\mathbb{E}\left[\exp(u\varepsilon_{t+1})\,|\,\underline{w_t},\,,\underline{z_t}\right] \\
= \;& \exp\{-u[\nu(Z_t)\mu(W_t, Z_t) + \varphi(Z_t)' W_t] \\
& \qquad + A[u; \varphi(Z_t), \mu(W_t, Z_t)]' W_t + \; b[u; \nu(Z_t), \mu(W_t, Z_t)]\} \\
= \;& \exp\{[A[u; \varphi(Z_t), \mu(W_t, Z_t)] - u\varphi(Z_t)]' W_t \\
& \qquad + b[u; \nu(Z_t), \mu(W_t, Z_t)] - u\,\nu(Z_t)\,\mu(W_t, Z_t)\}\;.
\end{aligned}
\tag{III.4}
$$

- Note that the dynamics of (w_t, z_t) is in general not Car.

III.2 – SWITCHING REGIMES ARG ATSMs

The Stochastic Discount Factor

- In the $SwARG(p)$ model the SDF is specified in the following way:

$$\begin{aligned} & M_{t,t+1} \\ = \quad & \exp\Big\{-c'W_t - d'Z_t + \Gamma(Z_t,W_t)\,\varepsilon_{t+1} + \Gamma(Z_t,W_t)\,[\nu(Z_t)\mu(W_t,Z_t) + \varphi(Z_t)'W_t] \\ & \quad - A[\Gamma(Z_t,W_t);\varphi(Z_t),\mu(W_t,Z_t)]'W_t \\ & \quad\quad - b[\Gamma(Z_t,W_t);\nu(Z_t),\mu(W_t,Z_t)] - \delta(Z_t,W_t)'z_{t+1}\Big\}\,, \end{aligned} \tag{III.5}$$

where $\Gamma(Z_t,W_t) = \gamma(Z_t) + \tilde{\gamma}'(Z_t)\,W_t$, or, equivalently,

$$\begin{aligned} M_{t,t+1} \quad = \quad & \exp\Big\{-c'W_t - d'Z_t + \Gamma(Z_t,W_t)\,w_{t+1} - A[\Gamma(Z_t,W_t);\varphi(Z_t),\mu(W_t,Z_t)]'W_t \\ & \quad - b[\Gamma(Z_t,W_t);\nu(Z_t),\mu(W_t,Z_t)] - \delta(Z_t,W_t)'z_{t+1}\Big\}\,. \end{aligned} \tag{III.6}$$

- Under the AAO, we get:

$$\begin{aligned} & \exp(-r_t) = \mathbb{E}_t\left(M_{t,t+1}\right) \\ = \quad & \exp\left[-c'W_t - d'Z_t\right] \times \textstyle\sum_{j=1}^{J} \pi\left(e_i, e_j\,;W_t\right)\exp\left[-\delta(Z_t,W_t)'e_j\right]\,. \end{aligned}$$

III.2 – SWITCHING REGIMES ARG ATSMs

Two Useful Lemmas

- Assuming the normalization condition:

$$\sum_{j=1}^{J} \pi\left(e_i, e_j\,; W_t\right) \exp\left[-\delta(Z_t, W_t)' e_j\right] = 1 \quad \forall Z_t, W_t\,, \qquad \text{(III.7)}$$

  we obtain:

$$r_t = c' W_t + d' Z_t\,. \qquad \text{(III.8)}$$

- In the subsequent sections we will use several times the following lemmas. Let us consider the functions:

$$\tilde{a}(u;\rho,\mu) = \frac{\rho u}{1-u\mu} \quad \text{and} \quad \tilde{b}(u;\nu,\mu) = -\nu\log(1-u\mu)\,.$$

- We have **Lemma 1**:

$$\tilde{a}(u+\alpha;\rho,\mu) - \tilde{a}(\alpha;\rho,\mu) = \tilde{a}(u;\rho^*,\mu^*)$$

$$\tilde{b}(u+\alpha;\nu,\mu) - \tilde{b}(\alpha;\nu,\mu) = \tilde{b}(u;\nu,\mu^*)$$

$$\text{with } \rho^* = \frac{\rho}{(1-\alpha\mu)^2},\ \ \mu^* = \frac{\mu}{1-\alpha\mu}\,,$$

- Lemma 1 immediately implies **Lemma 2**:

$$A[u+\alpha;\varphi(Z_t),\mu(W_t,Z_t)] - A[\alpha;\varphi(Z_t),\mu(W_t,Z_t)]$$

$$= A[u;\varphi^*(Z_t),\mu^*(W_t,Z_t)]$$

$$b[u+\alpha;\nu(Z_t),\mu(W_t,Z_t)] - b[\alpha;\nu(Z_t),\mu(W_t,Z_t)]$$

$$= b[u;\nu(Z_t),\mu^*(Z_t,W_t)]\,,$$

- with $\varphi^*(Z_t) = \dfrac{\varphi(Z_t)}{[1-\alpha\mu(Z_t,W_t)]^2}$
- and $\mu^*(Z_t,W_t) = \dfrac{\mu(W_t,Z_t)}{1-\alpha\mu(W_t,Z_t)}$.

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- The Laplace transform of the risk-neutral conditional distribution of (w_{t+1}, z_{t+1}) is, using the notation $\Gamma_t = \Gamma(W_t, Z_t)$:

$$
\begin{aligned}
& \mathbb{E}_t^{\mathbb{Q}}[\exp(u w_{t+1} + v' z_{t+1})] \\
= \quad & \mathbb{E}_t\{\exp[(u + \Gamma_t)\, w_{t+1} - A[\Gamma_t; \varphi(Z_t), \mu(W_t, Z_t)]' W_t \\
& \qquad - b[Z_t; \nu(Z_t), \mu(W_t, Z_t)] + (v - \delta(W_t, Z_t))' z_{t+1}]\} \\
= \quad & \exp\{[(A[u + \Gamma_t; \varphi(Z_t), \mu(W_t, Z_t)] - A[\Gamma_t; \varphi(Z_t), \mu(W_t, Z_t)])' W_t \\
& \qquad + b[u + \Gamma_t; \nu(Z_t), \mu(W_t, Z_t)] - b[\Gamma_t; \nu(Z_t), \mu(W_t, Z_t)]]\} \\
& \qquad \times \textstyle\sum_{j=1}^{J} \pi(z_t, e_j\, ; W_t) \exp[(v - \delta(Z_t, W_t))' e_j] \;,
\end{aligned}
\tag{III.9}
$$

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- and, using **Lemma 2**, (III.10) can be written:

$$\begin{aligned} & \mathbb{E}_t^{\mathbb{Q}}[\exp(uw_{t+1} + v'z_{t+1})] \\ = \quad & \exp\{A[u; \varphi^*(Z_t), \mu^*(W_t, Z_t)]' W_t + b[u; \nu(Z_t), \mu^*(Z_t, W_t)]\} \\ & \quad \times \textstyle\sum_{j=1}^{J} \pi(z_t, e_j\,; W_t) \exp\left[(v - \delta(Z_t, W_t))' e_j\right] , \end{aligned} \tag{III.10}$$

with $\varphi^*(Z_t) = \dfrac{\varphi(Z_t)}{[1 - \Gamma_t \mu(Z_t, W_t)]^2}$, and $\mu^*(Z_t, W_t) = \dfrac{\mu(W_t, Z_t)}{1 - \Gamma_t \mu(W_t, Z_t)}$.

- So, from (III.2), we see that the $\mathbb{Q}$-dynamics of w_{t+1}, given $(\underline{w_t}, \underline{z_t})$, is in the same class as the historical one and obtained by replacing $\varphi(Z_t)$ with $\varphi^*(Z_t)$, and $\mu(X_t, Z_t)$ with $\mu^*(Z_t, X_t)$.

- In order to get a generalized linear term structure we impose that the risk-neutral dynamics is a Switching Regime ARG Car(p) process.

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- Using the results on regime-switching ARG processes, we see that:
  - $\varphi^*(Z_t)$ and $\mu^*(Z_t, W_t)$ must be constant,
  - $\nu(Z_t) = {\nu^*}' Z_t$
  - and $\pi\,(z_t, e_j\,; W_t) = \pi^*\,(z_t, e_j) \exp\left[(\delta(Z_t, W_t))' e_j\right]$;
- μ^* must be positive as well as the components of ν^* and φ^*.
- $\varphi(Z_t) = \dfrac{\varphi^*}{\mu^{*2}}\, \mu(W_t, Z_t)^2$, so $\mu(W_t, Z_t)$ must depend only on Z_t, and therefore the same is true for $\Gamma(W_t, Z_t)$.

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- Finally, we have the constraints:

  i) $\mu(Z_t) = \mu^*[1 - \Gamma(Z_t)\mu(Z_t)]$;

  ii) $\varphi(Z_t) = \varphi^*[1 - \Gamma(Z_t)\mu(Z_t)]^2$;

  iii) $\nu(Z_t) = \nu^{*'} Z_t$;

  iv) $\delta_j(W_t, Z_t) = \log \left[\dfrac{\pi\left(z_t, e_j \,;\, W_t\right)}{\pi^*(z_t, e_j)} \right]$.

- In particular, since $\varphi(Z_t) = \dfrac{\varphi^*}{\mu^{*2}} \mu(Z_t)^2$, the random vector must be proportional to a deterministic vector.

- Let us introduce the notations:

$$A^*(u) \quad = \quad A(u; \varphi^*, \mu^*)\,, \quad \tilde{C}_h = (C_{2,h}, \ldots, C_{p,h}, 0)'\,. \qquad \text{(III.11)}$$

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- In the univariate SwARG(p) ATSM the price at date t of the zero-coupon bond with residual maturity h is:

$$B(t,h) = \exp\left(C_h' \, W_t + D_h' \, Z_t\right) \,, \text{ for } h \geq 1 \,, \tag{III.12}$$

where the vectors C_h and D_h satisfy the following recursive equations:

$$\left\{ \begin{array}{lcl} C_h & = & -c + A^*(C_{1,h-1}) + \tilde{C}_{h-1} \\ D_h & = & -d \, - \, \nu^* \log(1 - C_{1,h-1}\mu^*) + \, \tilde{D}_{h-1} + F(D_{1,h-1}) \,, \end{array} \right. \tag{III.13}$$

where $C_{1,h-1} =$ first component of the p-dimensional vector C_{h-1},
$D_{h-1} = (D'_{1,h-1}, D'_{2,h-1})'$, $\tilde{D}_{h-1} = (D'_{2,h-1}, 0)'$,
$D_{1,h-1} =$ the first J-dimensional component of D_{h-1},
$D_{2,h-1} =$ the remaining $p\,J$-dimensional components of D_{h-1},
$F(D_{1,h-1}) = e_1 \otimes a_z(D_{1,h-1}, \pi^*)$,
e_1 being the vector $(1, 0, \ldots, 0)'$ of size $(p+1)$;
the initial conditions are $C_0 = 0$, $D_0 = 0$ (or $C_1 = -c$, $D_1 = -d$).

III.2 – SWITCHING REGIMES ARG ATSMs

The $\mathbb{Q}$-dynamics

- Again, we obtain a generalized linear term structure given by:

$$R(t,h) \quad = \quad -\frac{C_h'}{h} W_t - \frac{D_h'}{h} Z_t \,, \quad h \geq 1 \,, \qquad \text{(III.14)}$$

- Since $r_t = R(t,1) = c' W_t + d' Z_t$, and since the components of W_t are positive, the short term process will be positive as soon as the components of c and d are nonnegative.
- The positiveness of r_t implies that of $R(t,h)$, at any date t and time to maturity h, because $R(t,h) = -\frac{1}{h} \log \mathbb{E}_t^{\mathbb{Q}} [\exp(-r_t - \ldots \; -r_{t+h-1})]$.
- This positiveness can also be observed from the recursive equations generating the theoretical yields: using the fact that μ^* and the components of φ^* and ν^* are positive and that $0 < \pi_{ij}^* < 1$, it easily seen that, for any $u < 0$, the components of $A^*(u)$ and $-\nu^* \log(1 - C_{1,h-1}\mu^*)$ are negative and the result follows.

III.3 – WISHART AND QUADRATIC TSMs

Wishart QTSMs

- The Wishart Quadratic Term Structure model, proposed by Gourieroux and Sufana (2011, JEDC), is characterized by an unobservable factor (Y_t) which follows the Wishart autoregressive (WAR) process ($WAR_L(K, M, \Omega)$).

- The WAR process of order 1, called $WAR(1)$, is a Markov process Y_{t+1} valued in the space of $(L \times L)$ symmetric positive definite matrices, with conditional historical Laplace:

$$\begin{aligned}
& \mathbb{E}[\exp\ Tr(\Gamma Y_{t+1}) | \underline{Y_t}] \qquad \text{(III.15)} \\
= \quad & \exp\left\{ Tr[M'\Gamma(Id - 2\Omega\Gamma)^{-1} M Y_t] - \frac{K}{2} \log[det(Id - 2\Omega\Gamma)] \right\},
\end{aligned}$$

where Γ is a symmetric matrix; $Tr(\Gamma Y_{t+1})$ is:

$$\sum_{i=1}^{L} (\Gamma Y_{t+1})_{ii} = \sum_{i=1}^{L} \sum_{j=1}^{L} \gamma_{ij} Y_{t+1,ij} = \sum_{i=1}^{L} \gamma_{ii} Y_{t+1,ii} + \sum_{i<j}^{L} (\gamma_{ij} + \gamma_{ji}) Y_{t+1,ij}.$$

- K is the (positive) scalar degree of freedom ($K > L - 1$);

- M is a $(L \times L)$ matrix (of AR parameters),

- Ω is a $(L \times L)$ symmetric positive definite matrix.

III.3 – WISHART AND QUADRATIC ATSMs

Wishart QTSMs

- If K integer, Y_{t+1} obtained from:

$$\begin{cases} Y_{t+1} & = & \sum_{k=1}^{K} x_{k,t+1} x'_{k,t+1} \\ x_{k,t+1} & = & M x_{k,t} + \varepsilon_{k,t+1}, \quad k \in \{1, \dots, K\}, \end{cases}$$

where $\varepsilon_{k,t+1} \sim i.i.d.\mathcal{N}(0, \Omega)$ (independent across k's). We have:

$$Y_{t+1} = M\, Y_t\, M' + K\Omega, + \eta_{t+1}$$

where η_{t+1} is a matrix martingale difference, i.e. Y_t follows a matrix semi-strong AR(1)-like process.

- Without loss of generality (for identification reasons), we may assume $\Omega = I_L$. Particular case: $L = 1$ (univariate).

$$\mathbb{E}[\exp(uY_{t+1}) \,|\, \underline{Y_t}] = \exp\left[\frac{um^2}{1 - 2\omega u} Y_t - \frac{K}{2} \log(1 - 2\omega u)\right]$$

$\Leftrightarrow$ $ARG(1)$ process with $\rho = m^2$, $\mu = 2\omega$, $\nu = \frac{K}{2}$.

III.3 – WISHART AND QUADRATIC ATSMs

Wishart QTSMs

- The SDF is defined by:

$$M_{t,t+1} = \exp\left[Tr(CY_{t+1}) + d\right] , \tag{III.16}$$

  where C is a $(L \times L)$ symmetric matrix and d is a scalar.

- The associated R.N. conditional log-Laplace transform is:

$$\begin{aligned}\psi_t^{\mathbb{Q}}(\Gamma) &= Tr\left[M' \left\{(C+\Gamma)[I_L - 2(C+\Gamma)]^{-1} - C(I_L - 2C)^{-1}\right\} M\, Y_t\right] \\ &\quad -\frac{K}{2}\text{Log det}[(I_L - 2(I_L - 2C)^{-1}\Gamma)] ,\end{aligned}$$

  which is also Car(1).

- The yield curve at date t is affine in Y_t:

$$\begin{aligned}R(t,h) &= -\frac{1}{h}Tr[A(h)\, Y_t] - \frac{1}{h}b(h) ,\ \ h \geq 1 \\ A(h) &= M'[C + A(h-1)]\left\{I_L - 2[C + A(h-1)]\right\}^{-1} M \\ b(h) &= d + b(h-1) - \tfrac{K}{2}\text{Log det}[I_L - 2(C + A(h-1))] \\ A(0) &= 0 ,\ \ b(0) = 0 .\end{aligned}$$

III.3 – WISHART AND QUADRATIC ATSMs

Wishart QTSMs

- If C is negative semi-definite:
  - $A(h) \ll 0$ and $Tr[A(h)\, Y_t] \leq 0$;
  - $A(h)$ decreasing;
  - support of $R(t,h)$ is $\left[-\frac{1}{h}\, b(h), +\infty\right]$, where $-\frac{1}{h}\, b(h) \geq 0$ and increasing with h; in particular, $R(t,h) \geq 0$ for all t and h and if $d = \frac{K}{2} \log \det(I_L - 2\,C)$, the support of $R(t,h)$ is $\mathbb{R}^+$.
- If K is integer, we get:

$$
\begin{aligned}
R(t,h) &= -\frac{1}{h}\, Tr[\sum_{k=1}^{K} A(h)\, x_{k,t}\, x'_{k,t}] - \frac{1}{h}\, b(h)\,, \;\; h \geq 1 \\
&= -\frac{1}{h} \sum_{k=1}^{K} x'_{k,t} A(h)\, x_{k,t} - \frac{1}{h}\, b(h)\,, \;\; h \geq 1\,,
\end{aligned}
\tag{III.17}
$$

  which is a sum of quadratic forms in $x_{k,t}$.
- If $K = 1$, then $R(t,h) = -\frac{1}{h} x'_t A(h)\, x_t - \frac{1}{h}\, b(h)\,, \;\; h \geq 1$ (standard Quadratic Term Structure Model (QTSM)).

III.3 – WISHART AND QUADRATIC ATSMs

Quadratic TSMs

- In the standard QTSM the process (x_t) has no constant term in the drift. We can also define a quadratic term structure model with a linear term, if the historical dynamics of x_{t+1} is given by the following Gaussian VAR(1) process:

$$\begin{aligned} x_{t+1} &= m + Mx_t + \varepsilon_{t+1}\,, \\ \varepsilon_{t+1} &\overset{\mathbb{P}}{\sim} IIN(0,\Sigma)\,. \end{aligned} \tag{III.18}$$

- Indeed (as seen in Lecture 1), the factor $w_t = [x_t', vech(x_t x_t')']'$ is Car(1), that is, w_t is an Extended Car process in the historical world. Moreover, choosing (with C_2 a symmetric (L,L) matrix):

$$\begin{aligned} M_{t,t+1} &= \exp\left[C_1' x_{t+1} + Tr(C_2 x_{t+1} x_{t+1}') + d\right] \\ &= \exp(C_1' x_{t+1} + x_{t+1}' C_2 x_{t+1} + d)\,, \end{aligned} \tag{III.19}$$

the process w_t is also Extended Car in the risk-neutral world. The term structure at date t is affine in w_t:

$$R(t,h) = x_t'\Lambda(h)\,x_t + \mu(h)'\,x_t + \nu(h)\,,\ \ h \geq 1\,, \tag{III.20}$$

where $\Lambda(h), \mu(h)$ and $\nu(h)$ follow recursive equations [see also Cheng and Scaillet (2007, MF)].

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

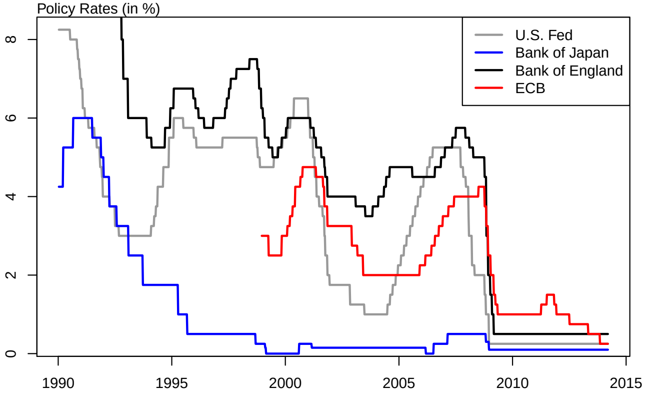
- In the presence of physical currency, absence of arbitrage opportunity and of storing cost of cash, nominal interest rates should be ≥ 0.
- Many standard models (e.g. Gaussian ATSM) are non consistent with non-negative nominal yields.
- **Shadow-rate approach** [Black (1995)]: The short term rate is given by:

$$r_t = \max(s_t, 0),$$

  where s_t is the shadow short-term interest rate. Pricing formula not available in closed-form. Approximation formula available [e.g. Krippner (2013), Priebsch (2013), Wu and Xia (2016)].
- Monfort, Pegoraro, Renne and Roussellet (2017) introduce an affine framework where the short-term rate can stay at zero for a prolonged period of time and with a stochastic lift-off probability.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

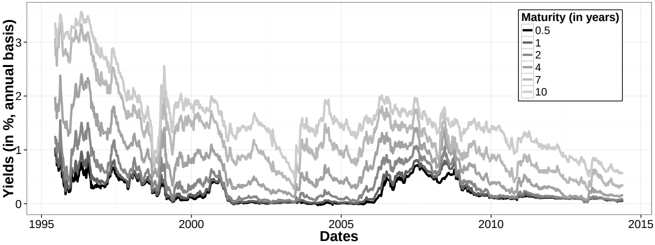
Zero lower bound (ZLB) issue



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Stylized facts to match

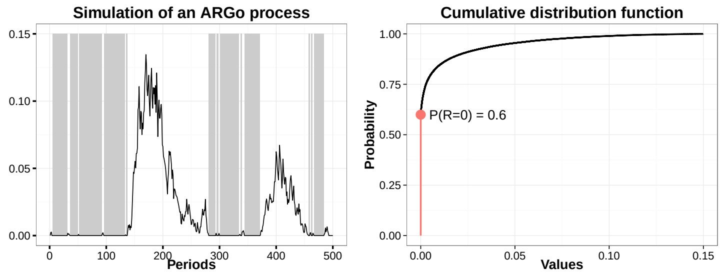
- The short-term nominal rate can stay at the ZLB for several periods...
- and in the meantime, longer-term yields can show substantial fluctuations [JGB yields from June 1995 to May 2014]



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

A new ZLB model: a primer

$\longrightarrow$ We introduce a new affine process:



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

What we do in this section

- **We derive** Non-negative Affine processes staying at 0 (ARG_0 processes) to build a Term Structure Model which is:
  - providing positive yields for all maturities;
  - consistent with the ZLB (a short-rate experiencing prolonged periods at 0) *WHILE* long-term rates still fluctuates;
  - affine: thus closed-form formulas for bond-pricing and lift-off probabilities are available.
- Empirical assessment on JGB yields (June 1995 to May 2014). **Good performance** of our model in terms of:
  - fitting yield levels and conditional variances;
  - calculating Risk-Neutral *and* Historical lift-off probabilities.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Related literature

- Term structure models at the ZLB: Black (1995), Ichiue & Ueno (2007), Kim & Singleton (2012), Krippner (2012), Renne (2012), Kim & Priebsch (2013), Wu & Xia (2013), Bauer & Rudebusch (2013), Christensen & Rudebusch (2013).
- Conditional volatilities of yields: Almeida *et al.* (2011), Bikbov & Chernov (2011), Filipovic, Larsson & Trolle (2013), Creal & Wu (2014), Christensen *et al.* (2014).
- Affine and Autoregressive Gamma processes: Darolles *et al.* (2006), Gourieroux & Jasiak (2006), Dai, Le & Singleton (2010), Creal & Wu (2013)
- Lift-off probabilities: Bauer & Rudebusch (2013), Swanson & Williams (2013)

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

The ARG_0 process

- Let us consider now the Gamma-Zero distribution $\gamma_0(\lambda,\mu)$. We construct this new distribution in two steps (introducing a mixing variable Z):
  1. $Z \sim \mathcal{P}(\lambda) \Longrightarrow Z(\omega) \in \{\mathbf{0}, 1, 2, \ldots\}$ and $\mathbb{P}(Z=0) = \exp(-\lambda)$.
  2. We define $W|Z \sim \gamma_Z(\mu)$, which implies:
     - If $Z=0$, W is a dirac point mass at 0.
     - If $Z \neq 0$, W is gamma-distributed (continuous on $\mathbb{R}^+$).

Definition

The non-negative r.v. $W \sim \gamma_0(\lambda,\mu)$, $\lambda > 0$ and $\mu > 0$, if

$$\begin{aligned} & W \,|\, Z \sim \gamma_Z(\mu) \quad \text{with} \quad Z \sim \mathcal{P}(\lambda) \\ \Rightarrow \qquad & \mathbb{P}(W=0) = \mathbb{P}(Z=0) = \exp(-\lambda)\,. \end{aligned}$$

- In other words, $W \sim \gamma_0(\lambda,\mu)$ if its (complicated) p.d.f. is:

$$f_W(w\,;\,\lambda,\mu) = \sum_{z=1}^{+\infty} \left[\frac{\exp(-w/\mu)\, w^{z-1}}{(z-1)!\,\mu^z} \times \frac{\exp(-\lambda)\lambda^z}{z\,!} \right] \mathbb{1}_{\{w>0\}} + \exp(-\lambda)\mathbb{1}_{\{w=0\}}$$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

The ARG$_0$ process

- However, it is characterized by a simple Laplace transform (LT):

$$\varphi_W(u\,;\,\lambda,\mu) := \mathbb{E}\left[\exp(uW)\right] = \exp\left[\lambda\frac{u\mu}{(1-u\mu)}\right] \quad \text{for} \quad u < \frac{1}{\mu}\,.$$

$\Longrightarrow$ which is exponential-affine in λ.

Definition

(w_t) is a ARG$_0(\alpha,\beta,\mu)$ if $(w_{t+1}|\underline{w_t})$ is Gamma-Zero distributed:

$$(w_{t+1}|\underline{w_t}) \sim \gamma_0(\alpha+\beta w_t,\mu) \quad \text{for} \quad \alpha \geq 0,\ \mu > 0,\ \beta > 0\,.$$

That is:

$$\frac{w_{t+1}}{\mu}\,|\,z_{t+1} \sim \gamma(z_{t+1}), \quad z_{t+1}\,|\,w_t \sim \mathcal{P}(\alpha+\beta w_t) \qquad \text{(III.21)}$$

Again, simple conditional LT, exponential-affine in w_t:

$$\begin{aligned} \varphi_{w,t}(u\,;\,\alpha,\beta,\mu) &:= \mathbb{E}_t\left[\exp(uw_{t+1})\right] \\ &= \exp\left[\frac{u\mu}{1-u\mu}(\alpha+\beta\,w_t)\right]\,, \quad \text{for} \quad u < \frac{1}{\mu}\,. \end{aligned}$$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

The ARG_0 process

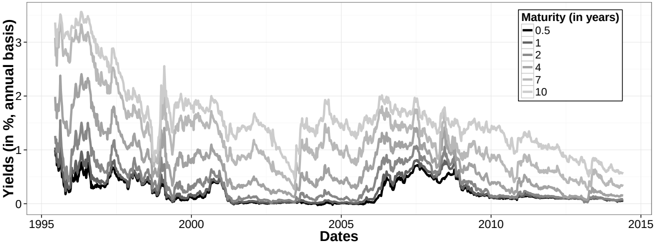
Key properties:

- Non-negative process.
- Affine process: the conditional Laplace transform is exp-affine.
- Staying at zero with probability:
  $$\mathbb{P}(w_{t+1} = 0 \mid w_t = 0) = \exp(-\alpha) \neq 0.$$
  - $\alpha \neq 0 \Longrightarrow$ zero is not absorbing.
  - in our multivariate yield curve model this probability will be time-varying, function of all date-t factors;
- Closed-form moments (affine conditional cumulants).
- General case: $w_{t+1} \mid w_t \sim ARG_\nu(\alpha, \beta, \mu)$.
  - $\frac{w_{t+1}}{\mu} \mid z_{t+1} \sim \gamma(\nu + z_{t+1})$ with $z_{t+1} \mid w_t \sim \mathcal{P}(\alpha + \beta\, w_t)$
  - $\nu = 0 \Longrightarrow ARG_0$ process.
  - $\nu > 0$, $\alpha = 0 \Longrightarrow$ ARG process of Gourieroux & Jasiak (2006).

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Stylized facts to match (1)

- short-term nominal rate at the ZLB for several periods
- longer-term yields showing substantial fluctuations [JGB yields from June 1995 to May 2014]



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Risk-neutral dynamics

- The state of the economy is defined by a n-dimensional vector w_t.

- These factors follow a $VARG_\nu$ process under $\mathbb{Q}$ (the same under $\mathbb{P}$).

$VARG_\nu$ processes

w_t follows a $\text{VARG}_\nu(\alpha, \beta, \mu)$ if, $\forall t$, $\forall i$:

- $z_{i,t+1} | w_t \sim \mathcal{P}(\alpha_i + \beta_i' w_t)$.

- $w_{i,t+1} | z_{i,t+1} \sim \gamma_{z_{i,t+1}+\nu_i}(\mu_i)$ cond. indep across i.

- Conditional $\mathbb{Q}$-moments (same formulas under $\mathbb{P}$):

$$
\begin{aligned}
\mathbb{E}_t^{\mathbb{Q}}(w_{t+1}) &= \mu^{\mathbb{Q}} \odot (\alpha^{\mathbb{Q}} + \beta^{\mathbb{Q}'} w_t + \nu) \\
\mathbb{V}_t^{\mathbb{Q}}(w_{t+1}) &= \text{diag}\left[\mu^{\mathbb{Q}} \odot \mu^{\mathbb{Q}} \odot \left(\nu + 2\alpha^{\mathbb{Q}} + 2\beta^{\mathbb{Q}'} w_t\right)\right]
\end{aligned}
$$

Note: Conditional correlations can be allowed.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Risk-neutral dynamics

- The vector of factors w_t is split into two: $w_t = (w_t^{(1)'}, w_t^{(2)'})'$ where:

  (i) All components of $w_t^{(1)}$ have $\nu_j = 0$ (point mass at 0).

  (ii) All components of $w_t^{(2)}$ have $\nu_j > 0$ (no point mass).

  (iii) $\mu_j^{\mathbb{P}} = 1$, $\beta^{\mathbb{P}}$ and $\beta^{\mathbb{Q}}$ lower-triangular (identification).

- The short-term rate r_t is given by:

$$\boxed{r_t = \delta' w_t^{(1)}} \quad (= r_{min} + \delta' w_t^{(1)}\,, \;\; \textit{if } LB \neq 0) \tag{III.22}$$

Key Property

$\{\text{Eq.(III.22)} + (i)\} \Rightarrow r_t$ has a zero point mass.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Risk-neutral dynamics

{Eq.(III.22) + *(iii)*}:

$$
\begin{aligned}
r_t &= \delta' w_t^{(1)} \\
\begin{pmatrix} w_t^{(1)} \\ w_t^{(2)} \end{pmatrix} &= \text{constant} + \begin{pmatrix} \beta_{11}^{\mathbb{Q}} & \beta_{12}^{\mathbb{Q}} \\ 0 & \beta_{22}^{\mathbb{Q}} \end{pmatrix} \begin{pmatrix} w_{t-1}^{(1)} \\ w_{t-1}^{(2)} \end{pmatrix} + \xi_t^{\mathbb{Q}}
\end{aligned}
$$

- We have $w^{(2)} \overset{\text{G.C.}}{\longrightarrow} w^{(1)}$
- and thus $w_t^{(2)}$ appears in the short rate conditional $\mathbb{Q}$-expectations (hence in long rates).

$\Longrightarrow$ long-term yields can move during the ZLB.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Pricing Formulas

- The model belongs to the class of ATSM.
- Explicit closed-form bond-pricing.
- Yields are affine in the factors for all maturities:

$$R(t,h) = -\frac{1}{h}\left(A_h{}' w_t + B_h\right) = \overline{A}_h' w_t + \overline{B}_h.$$

- Recursive pricing formulas:

$$\begin{aligned}
A_h &= -\delta + \beta^{\mathbb{Q}}\left(\frac{A_{h-1}\odot\mu^{\mathbb{Q}}}{1 - A_{h-1}\odot\mu^{\mathbb{Q}}}\right)\\
B_h &= B_{h-1} + \alpha^{\mathbb{Q}\prime}\left(\frac{A_{h-1}\odot\mu^{\mathbb{Q}}}{1 - A_{h-1}\odot\mu^{\mathbb{Q}}}\right) - \nu' \log\left(1 - A_{h-1}\odot\mu^{\mathbb{Q}}\right)
\end{aligned}$$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

The Historical dynamics

- The SDF is exp-affine with market price of risk vector θ:

$$\frac{d\mathbb{P}_{t,t+1}}{d\mathbb{Q}_{t,t+1}} = \exp\left[\theta' w_{t+1} - \psi_t^{\mathbb{Q}}(\theta)\right]$$

Change of measure property

w_t follows a $\text{VARG}_\nu(\alpha^{\mathbb{P}}, \beta^{\mathbb{P}}, \mu^{\mathbb{P}})$ process under the historical measure $\mathbb{P}$.

$$\alpha_j^{\mathbb{P}} = \frac{\alpha_j^{\mathbb{Q}}}{1-\theta_j\,\mu_j^{\mathbb{Q}}}\,, \qquad \beta_j^{\mathbb{P}} = \frac{1}{1-\theta_j\,\mu_j^{\mathbb{Q}}}\,\beta_j^{\mathbb{Q}}\,, \qquad \mu_j^{\mathbb{P}} = \frac{\mu_j^{\mathbb{Q}}}{1-\theta_j\,\mu_j^{\mathbb{Q}}}\,.$$

Rk: ν is the same under both measures.

- Conditional variance of yields:

$$\mathbb{V}_t^{\mathbb{P}}\left[R_{t+1}(h)\right] = \overline{A}_h'\mathbb{V}_t^{\mathbb{P}}(w_{t+1})\overline{A}_h$$

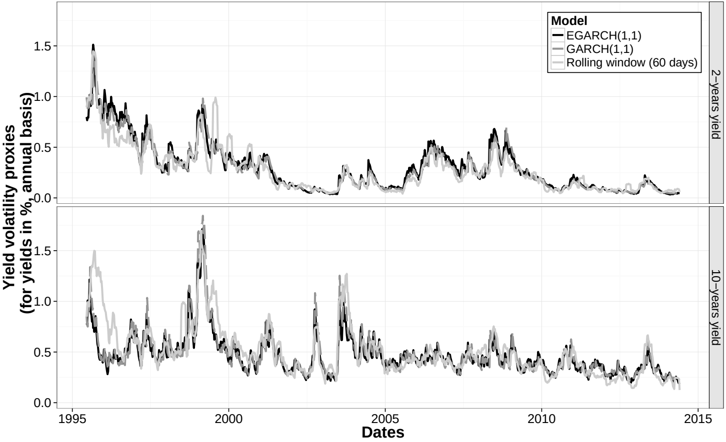
$$= \quad \overline{A}_h{}'\left\{\text{diag}\left[\mu^{\mathbb{P}} \odot \mu^{\mathbb{P}} \odot \left(\nu + 2\alpha^{\mathbb{P}} + 2\beta^{\mathbb{P}'} w_t\right)\right]\right\}\overline{A}_h$$

- Time-varying and maturity-dependent.

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Stylized facts to match (2)

Conditional volatilities: time-varying and maturity-dependent.



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Advantages of an affine framework

NATSM properties

- Yields $R(t,h)$ are non-negative;
- Long-term yields can move while $r_t = 0$ for several periods;
- Unconditional first two moments are available in closed-form;
- Conditional first two moments of yields are affine in w_t (available in closed-form);
- Yields forecasts are explicitly affine in w_t;

Estimation using State-Space formulation

State vector $Y_t = (R_t', V_t', S_t')'$ affine in w_t:

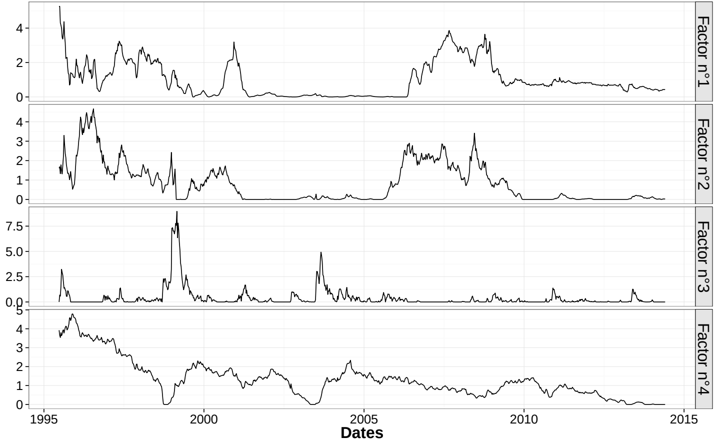
- R_t = yield levels (6 maturities);
- V_t = 2- and 10-y yield conditional (EGARCH) variance;
- S_t = SPF for 3-m and 1-y ahead 10-y yield;
- prelim. estimations suggests $\dim(w_t^{(1)}) = 1$, $\dim(w_t^{(2)}) = 3$, $\nu = 0$;
- Linear Kalman-filter-based QML:

$$\left\{ \begin{array}{lcl} w_{t+1} & = & m + Mw_t + \Sigma_t^{1/2}\varepsilon_{t+1} \\ Y_t & = & \Gamma_0 + \Gamma_1 w_t + \Omega\,\eta_t \end{array} \right. ,$$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Estimation results

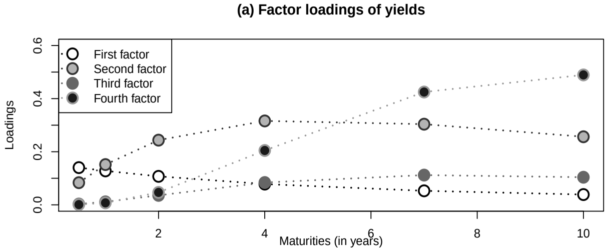
Filtered Factors

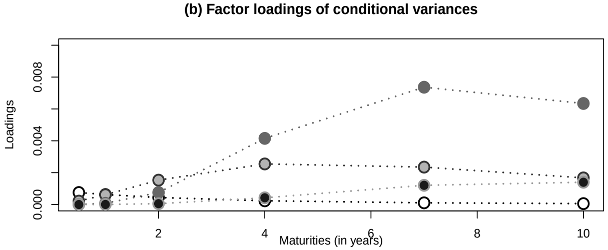


III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Estimation results

Factor loadings of yields and conditional variances.

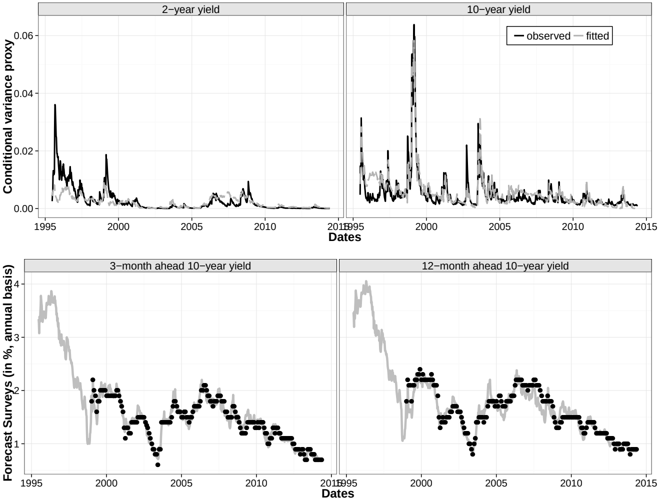




III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Estimation results

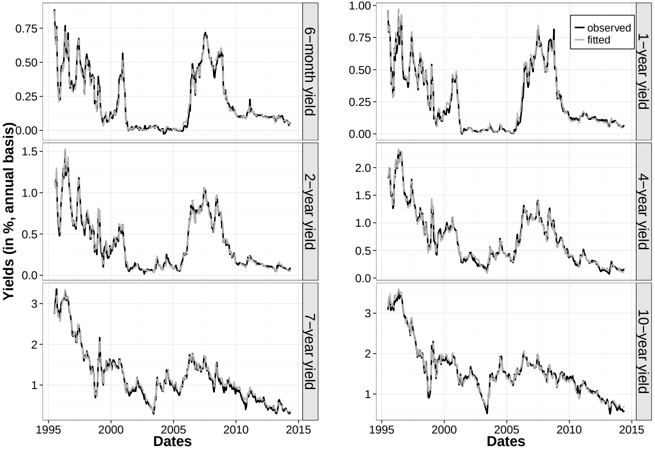
Fit of Conditional Variances and SPFs.



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Estimation results

Fit of Yields.



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Lift-offs formulas: univariate case

Useful formula

If $z \in \mathbb{R}^+$ and $\varphi_z(u)$ its Laplace transform.

$$\mathbb{P}(z=0) = \lim_{u \to -\infty} \varphi_z(u)\,.$$

- Lift-off probabilities: $(w_t) \sim \text{ARG}_0(\alpha, \beta, \mu)$ and $\varphi_{t,h}(u_1, \ldots, u_h)$ its multi-horizon conditional Laplace transform.
  - $\mathbb{P}(w_{t+h} = 0 \,|\, w_t) = \lim_{u \to -\infty} \varphi_{t,h}(0, \ldots, 0, u)$
  - $$\mathbb{P}\left[w_{t+1} = 0, \ldots, w_{t+h} = 0 \,\middle|\, w_t\right] = \lim_{u \to -\infty} \varphi_{t,h}(u, \ldots, u) = \exp(-\alpha\, h - \beta\, w_t)\,,$$
  - $$\mathbb{P}\left[w_{t+1} = 0, \ldots,\ w_{t+h} = 0, w_{t+h+1} > 0 \,\middle|\, w_t)\right] = \exp\left[-\alpha\, h - \beta\, w_t\right]\left[1 - \exp(-\alpha)\right]\,, \quad h > 1.$$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Lift-offs formulas: multivariate case

Useful formula

If $z \in \mathbb{R}^n_+$ and $\varphi_Z(u_1, \ldots, u_n)$ its Laplace transform.

$$\mathbb{P}_Z\{0, \ldots, 0\} = \lim_{u \to -\infty} \varphi_Z(u, \ldots, u)\,.$$

- Notation and formulas: Multi-horizon conditional LT.

$$\begin{aligned}
\varphi^{\mathbb{P}}_{t,k}(u_1, \ldots, u_k) &= \mathbb{E}^{\mathbb{P}}\left[\exp\left(u_1^{'} w_{t+1} + \ldots + u_k^{'} w_{t+k}\right) \,\middle|\, w_t\right] \\
&= \exp\left[\mathcal{A}_k^{'} X_t + \mathcal{B}_k\right]
\end{aligned}$$

$$\varphi^{(h)\mathbb{P}}_{R,t,k}(v_1, \ldots, v_k) = \mathbb{E}\left[\exp\left(v_1 R_{t+1}(h) + \ldots + v_k R_{t+k}(h)\right) \mid w_t\right]$$

- $\mathbb{P}\left[r_{t+k} = 0 \mid w_t\right] = \lim_{u \to -\infty} \varphi^{(1)\mathbb{P}}_{R,t,k}(0, \ldots, 0, u)$
- $\mathbb{P}\left[r_{t+1} = 0, \ldots, r_{t+k} = 0 \mid X_t\right]$
  $= \lim_{u \to -\infty} \varphi^{(1)\mathbb{P}}_{R,t,k}(u, \ldots, u) = p_{r,t,k} \quad \text{(say)}$

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Lift-off probability dates under $\mathbb{P}$ and $\mathbb{Q}$

- Notation and formulas:
- $\mathbb{P}\left[r_{t+1} = 0, \ldots, r_{t+k-1} = 0, r_{t+k} > 0 \,|\, w_t\right] = p_{r,t,k-1} - p_{r,t,k}$
- $$\mathbb{P}\left[v' R_{t+1}^{(t+k)}(h) > \lambda \,|\, w_t\right] = \frac{1}{2} + \frac{1}{\pi} \int_0^{+\infty} \frac{Im\left[\varphi_{R,t,k}^{(h)\mathbb{P}}(i\, v\, x) \exp(-i\, \lambda\, x)\right]}{x}\, dx$$

In what follows we calculate the following probabilities:

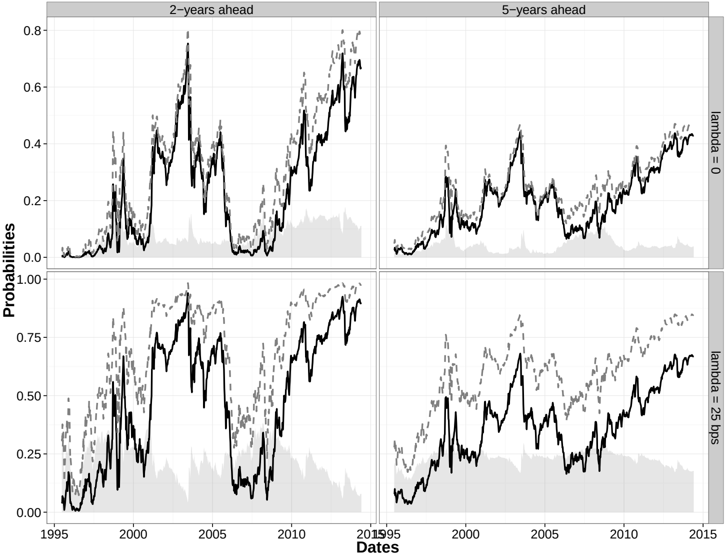
- $\mathbb{P}(r_{t+k} = 0 \,|\, \underline{w_t})$ and $\mathbb{Q}(r_{t+k} = 0 \,|\, \underline{w_t})$;
- $\mathbb{P}(r_{t+k} < 25\, bps. \,|\, \underline{w_t})$ and $\mathbb{Q}(r_{t+k} < 25\, bps. \,|\, \underline{w_t})$.

Next two plots ($\mathbb{Q}$ is the black solid line):

- *Time-series dimension*: t varies ($k = 2$yrs and 5yrs).
- *Horizon dimension*: k varies ($t = 11/30/07$ and $05/30/14$).

III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Lift-off probability dates under $\mathbb{P}$ and $\mathbb{Q}$



III.4 – NON-NEGATIVE AFFINE TERM STRUCTURE MODEL

Model-implied distributions of future short-term rates